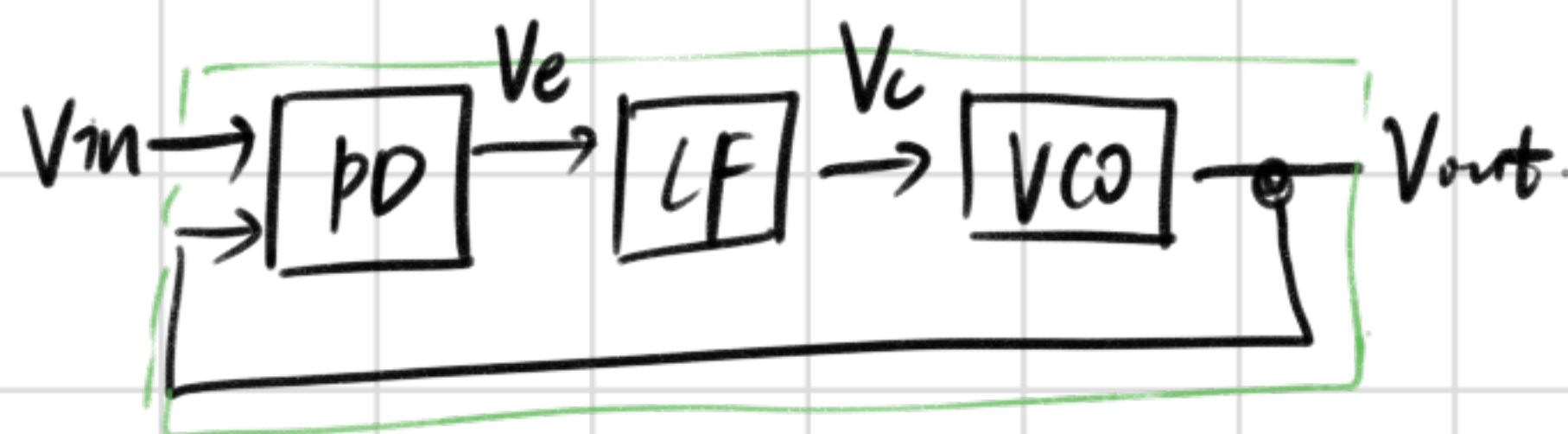


Basic operation of pll:



pll diagram

$$\begin{cases} V_{in} = A_{in} \sin(\omega_{in} t) & ; \quad \phi_{in}(t) = \omega_{in} t \\ V_{out} = A_{out} \sin(\omega_{out} \cdot t) & ; \quad \phi_{out}(t) = \omega_{out} \cdot t \\ \phi_{er}(t) = \phi_{in}(t) - \phi_{out}(t) \end{cases}$$

— ϕ_{out} follows ϕ_{in} ★

— Phase / Frequency

— TF: $\frac{\phi_{out}(s)}{\phi_{in}(s)}$

PD output: $V_e \propto \phi_{er}$

Loop filter filters error voltage to give V_c .

$\omega_{out} \propto V_c$

PD operation:



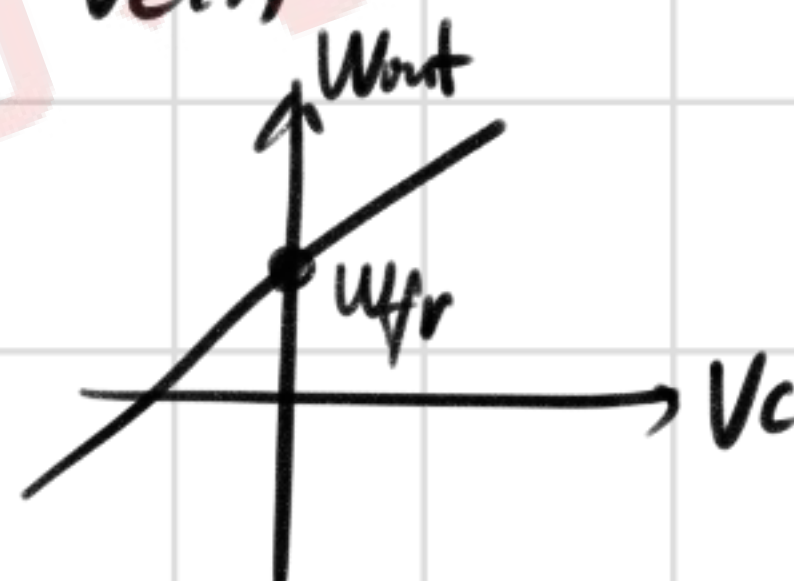
Gain of PD: $K_{PD} = \frac{dV_e}{d\phi_{er}} \text{ (V/rad)}$

Loop filter Transfer Function, $LF(s) = \frac{V_c(s)}{V_e(s)}$

$$\omega_{out} = \omega_{fr} + K_{VCO} \cdot V_c$$

free running frequency

gain of oscillator



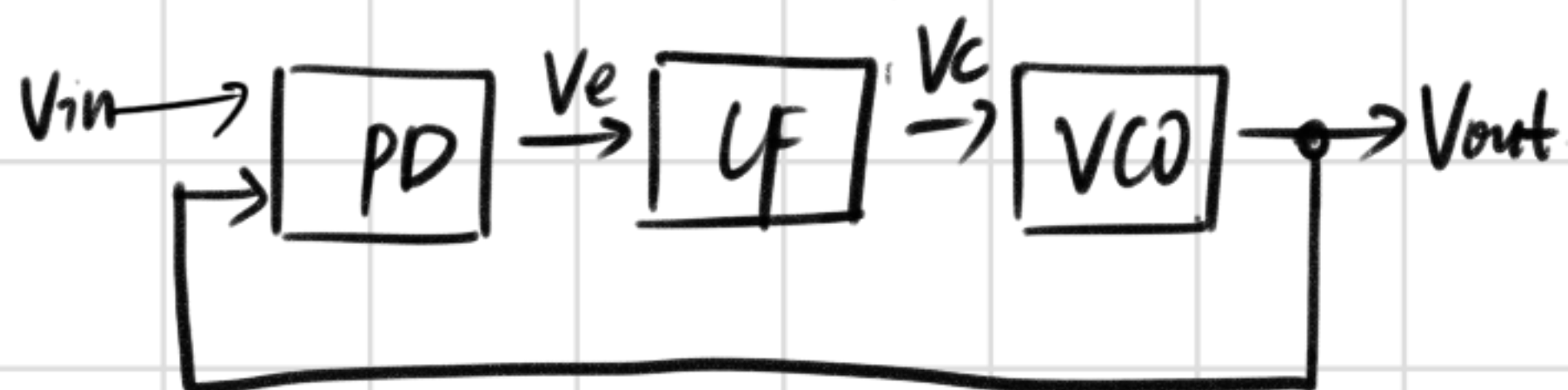
In steady state:

$$\omega_{in} = \omega_{out}$$

$$\frac{d\phi_{er}}{dt} = 0$$

> locked state

Simple implementation of PLL

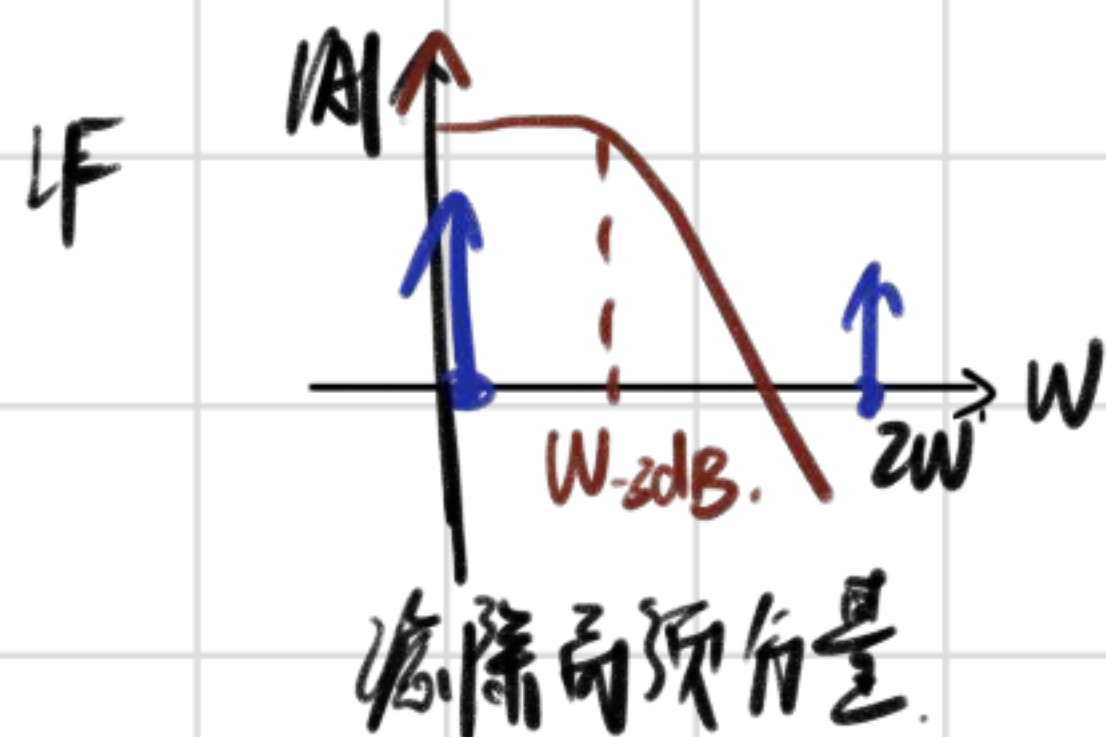
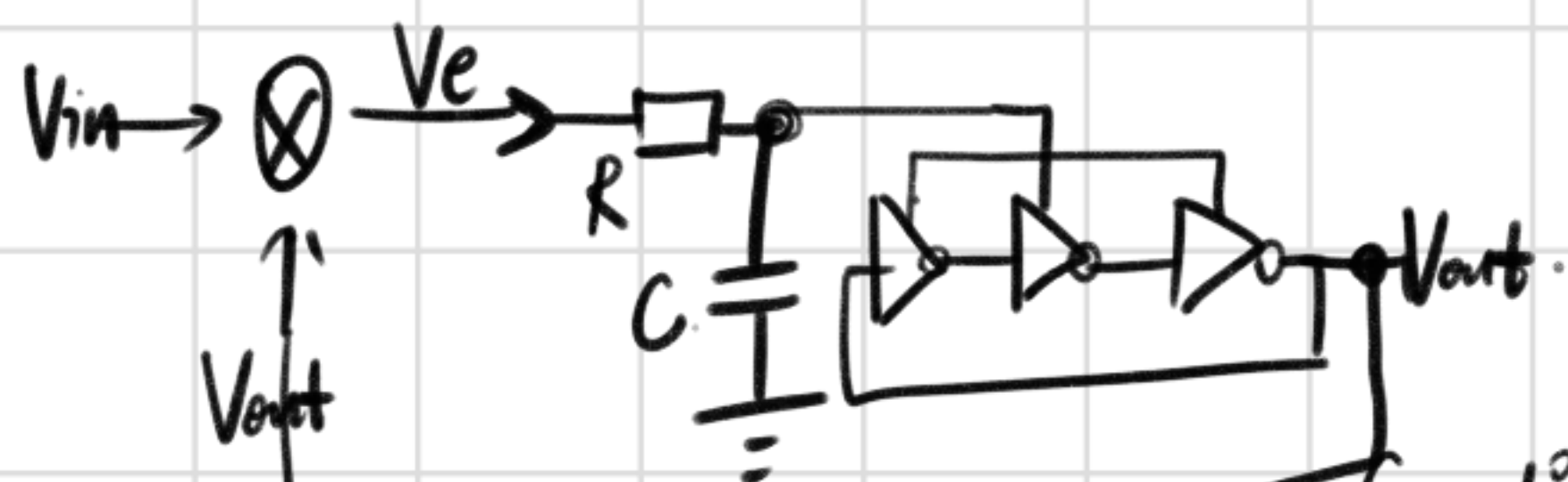


$$V_{in} = A_{in} \sin(\omega_{in} t + \phi_{in}(0))$$

$$V_{out} = A_{out} \cos(\omega_{out} t + \phi_{out}(0))$$

$$V_{in} \times V_{out} = A_{in} A_{out} \sin(\omega_{in} t) \cos(\omega_{out} t)$$

$$= \frac{A_{in} A_{out}}{2} [\sin(\overline{\omega_{in} + \omega_{out} t}) + \sin(\overline{\omega_{in} - \omega_{out} t})]$$



Case 1: $\omega_{in} = \omega_{out} = \omega$, $V_{in} \times V_{out} = \frac{A_{in} A_{out}}{2} [\sin(2\omega t) + 0]$

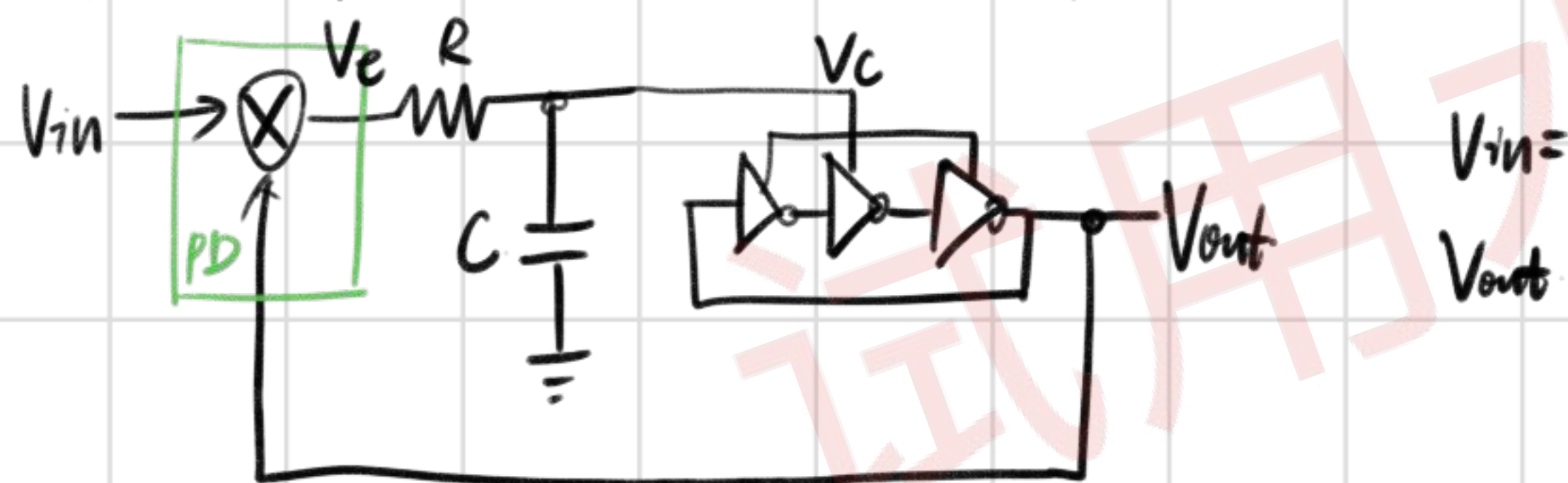
Case 2: $\omega_{in} = \omega$, $\omega_{out} = \omega - \Delta\omega$

$$V_{in} \times V_{out} = \frac{A_{in} A_{out}}{2} [\sin(2\omega - \Delta\omega t) + \sin(\Delta\omega t)]$$

Case 3: $\omega_{in} = \omega_{out} = \omega$, but $\phi_{in}(0) - \phi_{out}(0) = \phi_{er}(0)$

$$V_{in} \times V_{out} = \frac{A_{in} A_{out}}{2} [\sin(2\omega t + \phi_{in}(0) + \phi_{out}(0)) + \sin(\phi_{er}(0))]$$

Input-Output Characteristic of Basic PLL



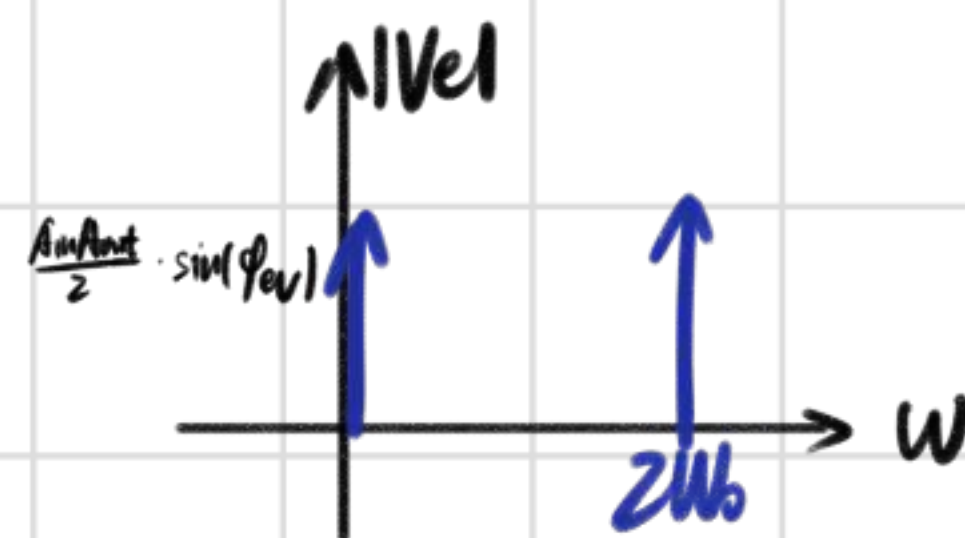
Phase detector:

$$V_e = \frac{A_{in} A_{out}}{2} [\sin(\overline{\omega_{in} + \omega_{out} t} + \phi_{in}(0) + \phi_{out}(0)) + \sin(\overline{\omega_{in} - \omega_{out} t} + \phi_{er}(0))]$$

In locked state of PLL, $\omega_{out} = \omega_{in} = \omega_0$, $\frac{d\phi_{er}}{dt} = 0$:

$$V_e(t) = \frac{A_{in} A_{out}}{2} [\sin(2\omega_0 t + \phi_{er}(0)) + \sin(\phi_{er}(0))]$$

$$\bar{V}_e = \frac{A_{in} A_{out}}{2} \sin(\phi_{er})$$



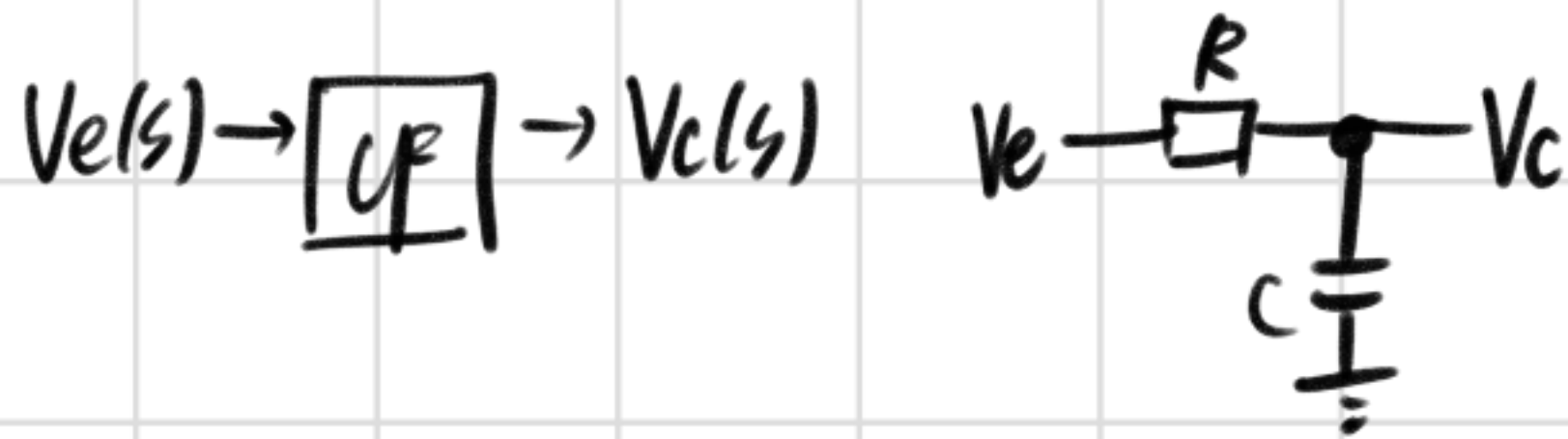
Gain of PD, $K_{PD} = \frac{d\bar{V}_e}{d\phi_{er}} = \frac{A_{in} A_{out}}{2} \cos(\phi_{er})$

K_{PD} 的值与环路锁定时 ϕ_{er} 有关

PD with frequency error at input: $\omega_{in} = \omega_0$, $\omega_{out} = \omega_0 - \Delta\omega$

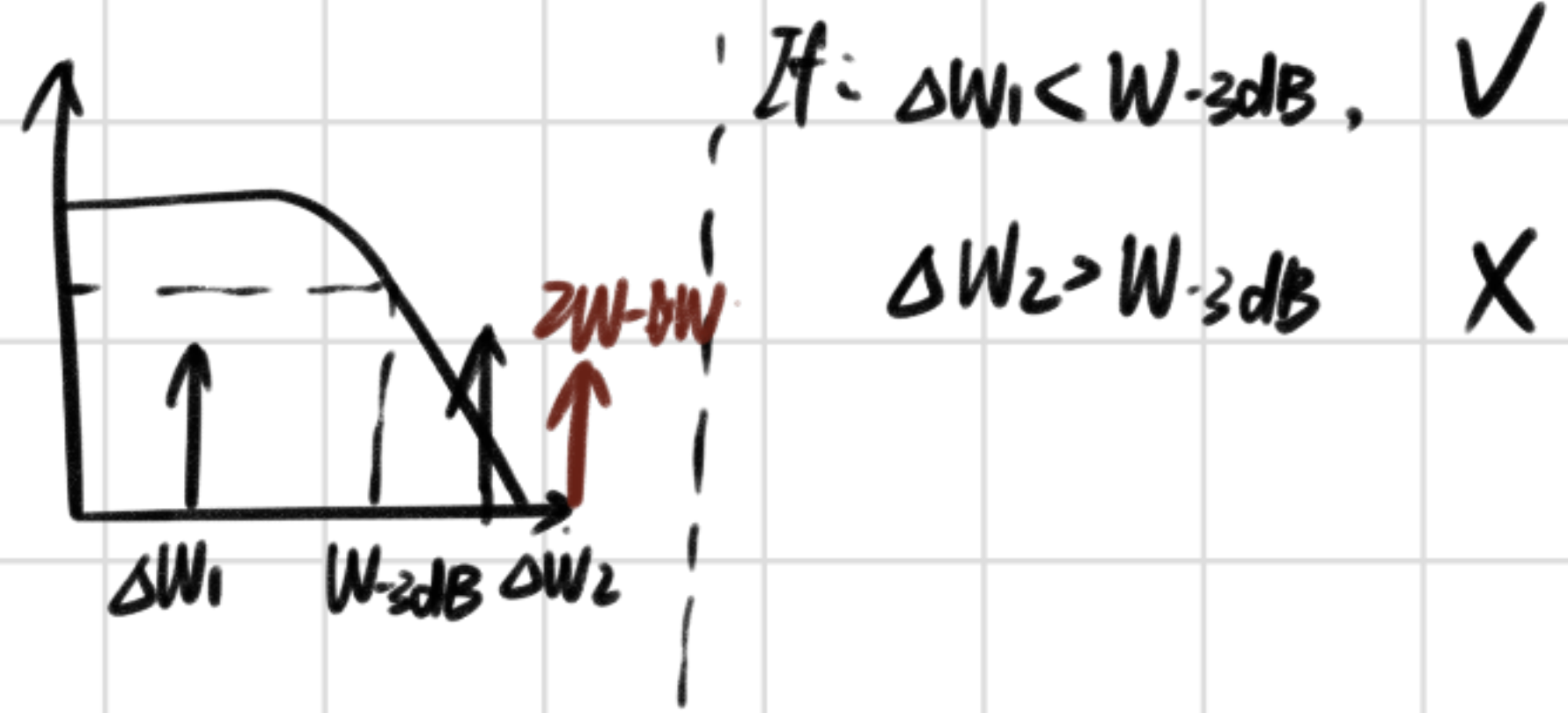
$$V_e = \frac{A_{in} A_{out}}{2} [\sin(\overline{2\omega - \Delta\omega} t) + \sin(\Delta\omega t)]$$

When considering **loop filter**,



$$|F(s)| = \frac{V_c(s)}{V_e(s)} = \frac{1}{1 + sRC}$$

$$\omega_{-3dB} = \frac{1}{RC}$$



How **VCO** output frequency changes:

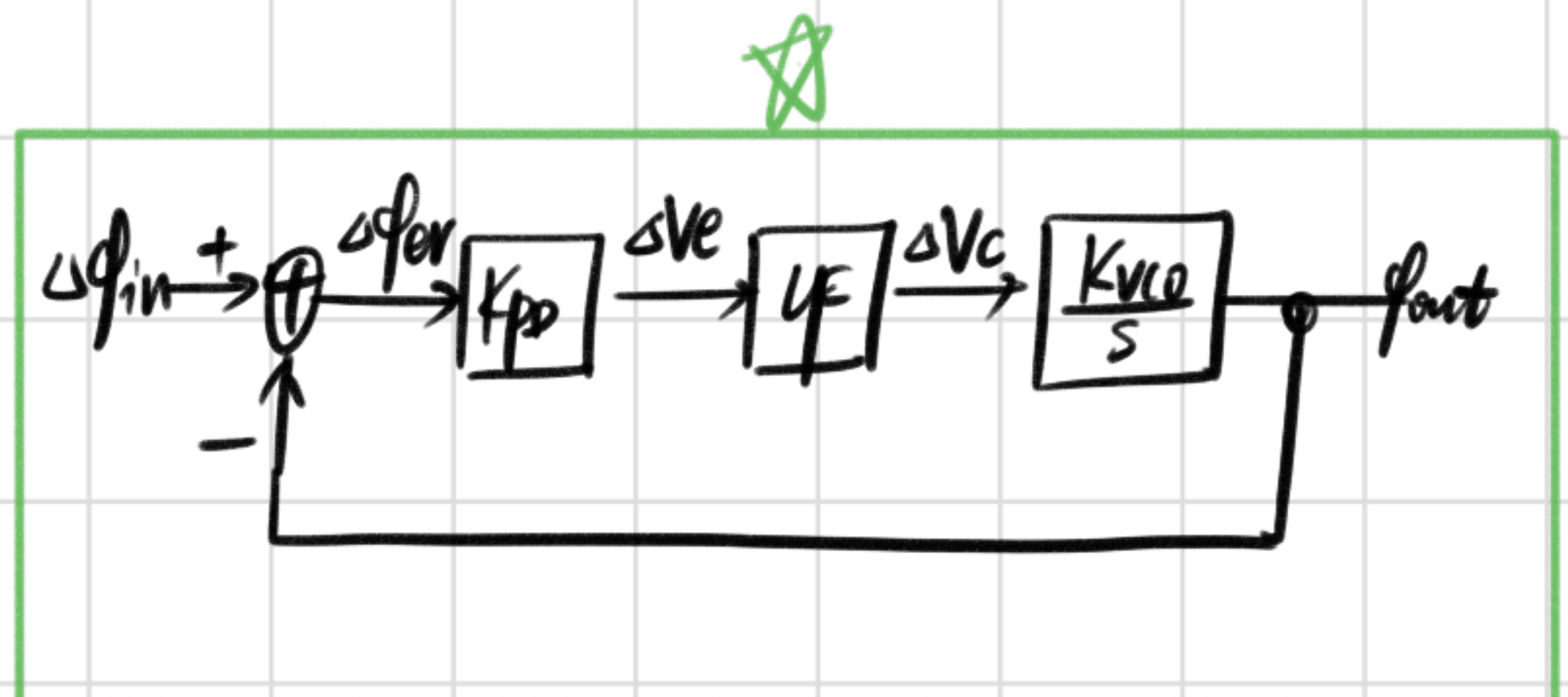
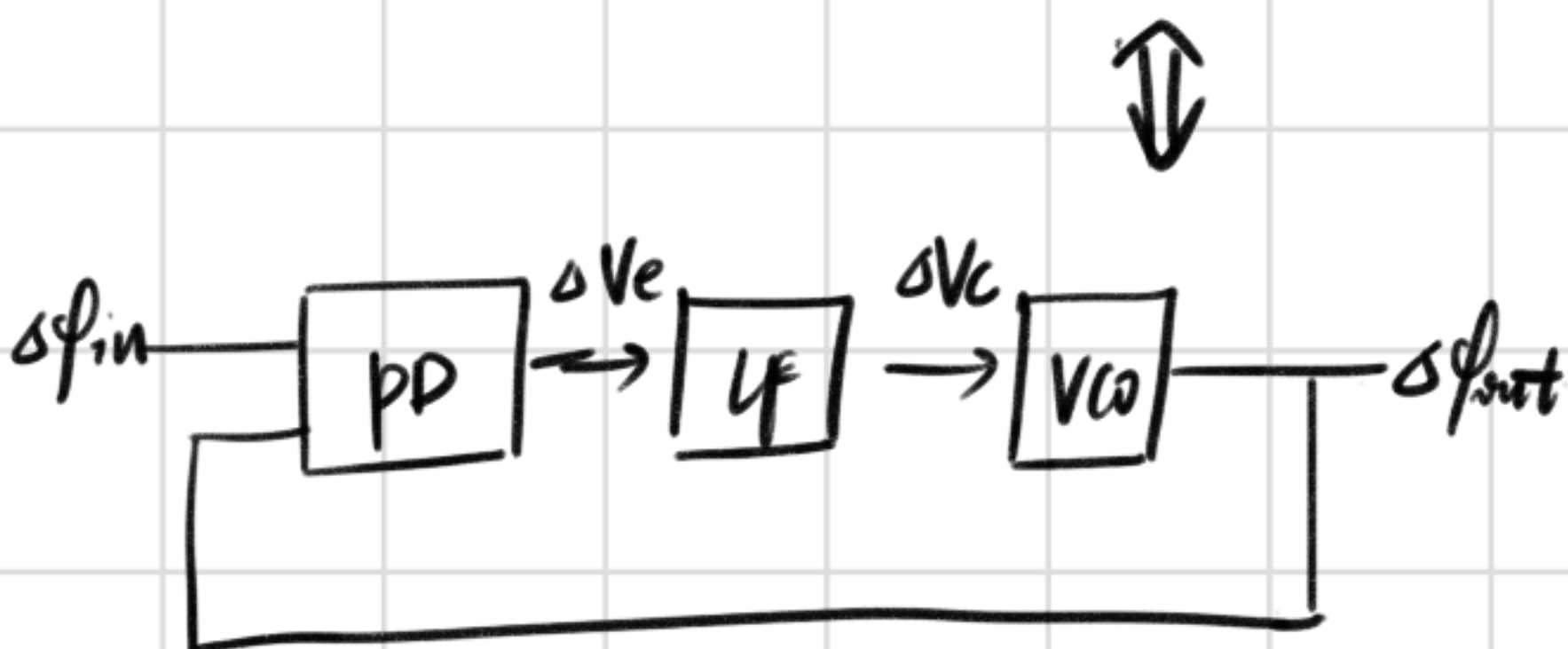
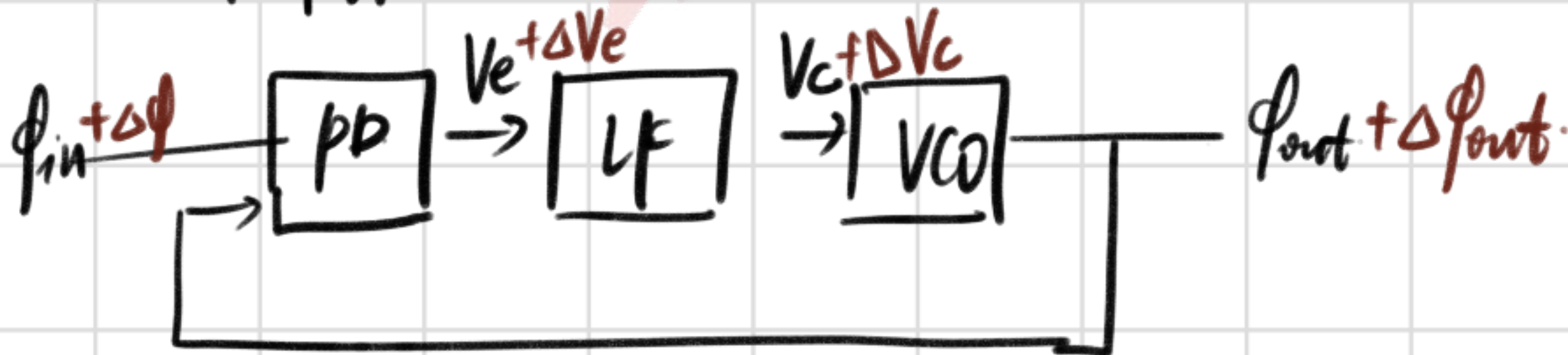
$$\omega_{out} = \omega_{fr} + K_{vco} \cdot V_c$$

$$\begin{aligned} \phi_{out} &= \int \omega_{out} \cdot dt = \int \omega_{fr} \cdot dt + K_{vco} \int V_c \cdot dt \\ &= \omega_{fr} \cdot t + K_{vco} \int V_c \cdot dt \end{aligned}$$

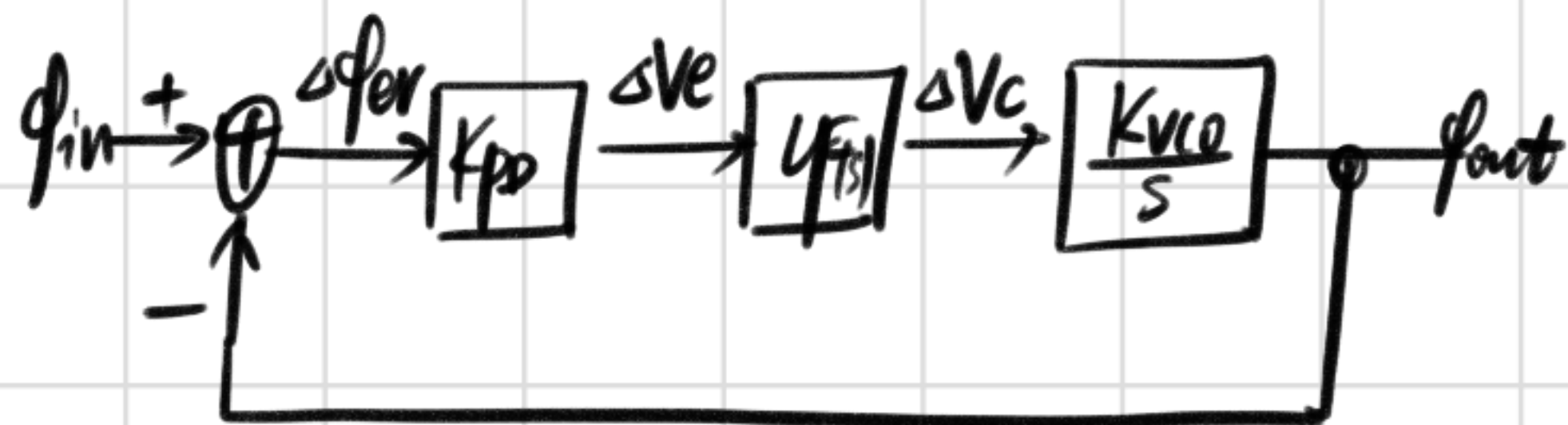
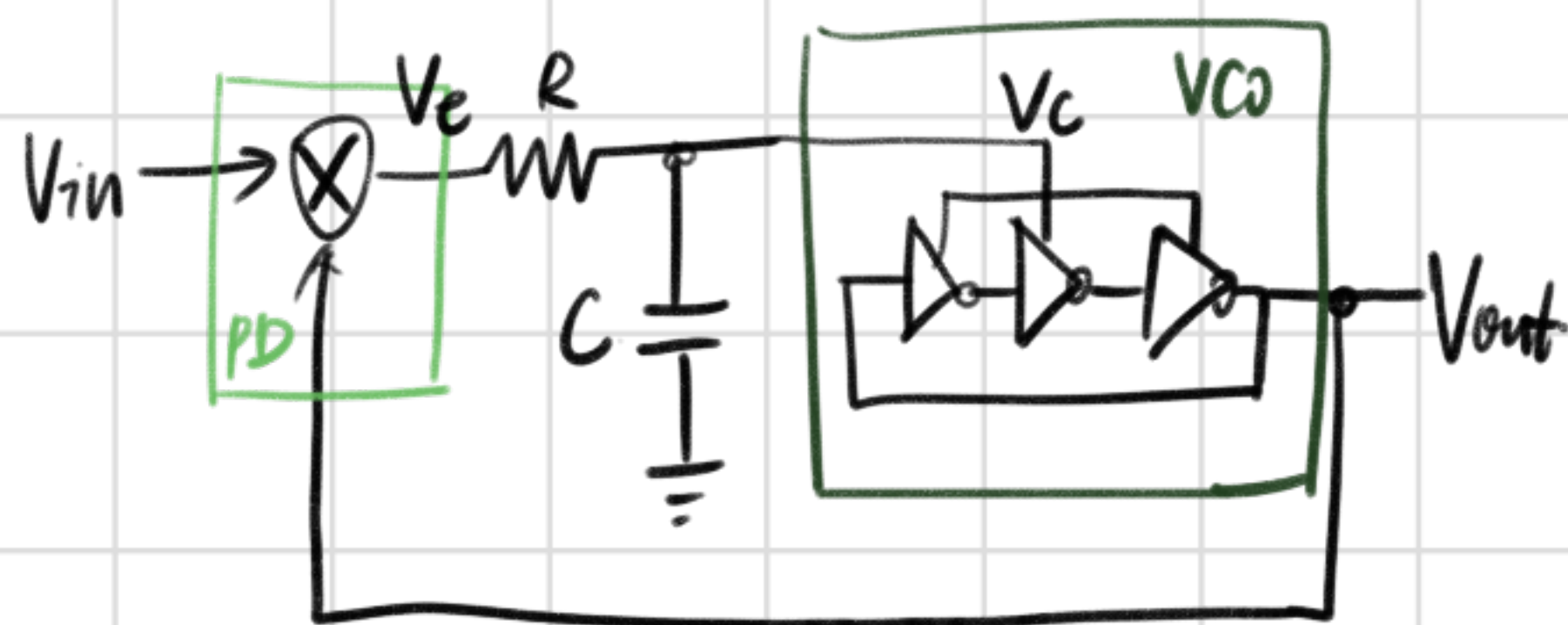
$$\Delta\phi_{out} = K_{vco} \int \Delta V_c \cdot dt \quad \text{In time domain}$$

$$\frac{\Delta\phi_{out}(s)}{\Delta V_c(s)} = \frac{K_{vco}}{s} \quad \text{--- VCO's transfer function}$$

In actual PLL:



Time Domain Analysis of a simple pll



Small-signal model of pll

ex: $V_{in} = 1 \cdot \sin(\omega_{in} t)$

$V_{out} = 1 \cdot \cos(\omega_{out} t)$

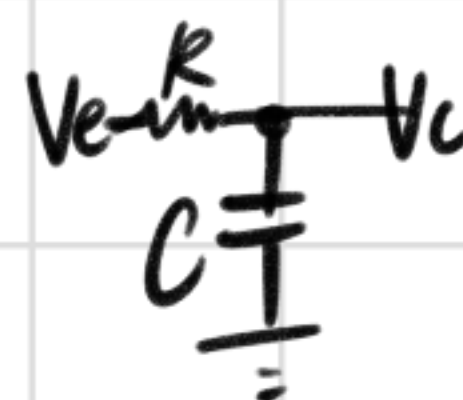
at $t=0$, $\omega_{in} = \omega$, $\omega_{out} = \omega - \Delta\omega(0) = \omega_{fr}$

$K_{vco} = 2\pi \times 100 \times 10^6 \text{ rad}$

$\omega = 2\pi \times 50 \text{ M} \cdot \text{rad/s}$

$\Delta\omega(0) = 2\pi \times 50 \text{ M} \cdot \text{rad/s}$

$V_{in} \rightarrow \phi \rightarrow V_e$
 $V_{out} \rightarrow \phi$
 $V_e = \frac{1}{2} [\sin(2\omega - \Delta\omega t) + \sin(\Delta\omega t)]$
 $\phi_{err} = \int \omega \cdot dt - \int (\omega - \Delta\omega(0)) dt = \Delta\omega(0) t$



$\Delta\omega(0) \ll \frac{1}{RC}$, $2\omega - \Delta\omega \gg \frac{1}{RC}$

$V_c(t) = \frac{1}{2} \sin(\Delta\omega(0) t)$

$V_c \rightarrow VCO \rightarrow V_{out}$ $\omega_{out} = \omega_{fr} + K_{vco} \cdot V_c$

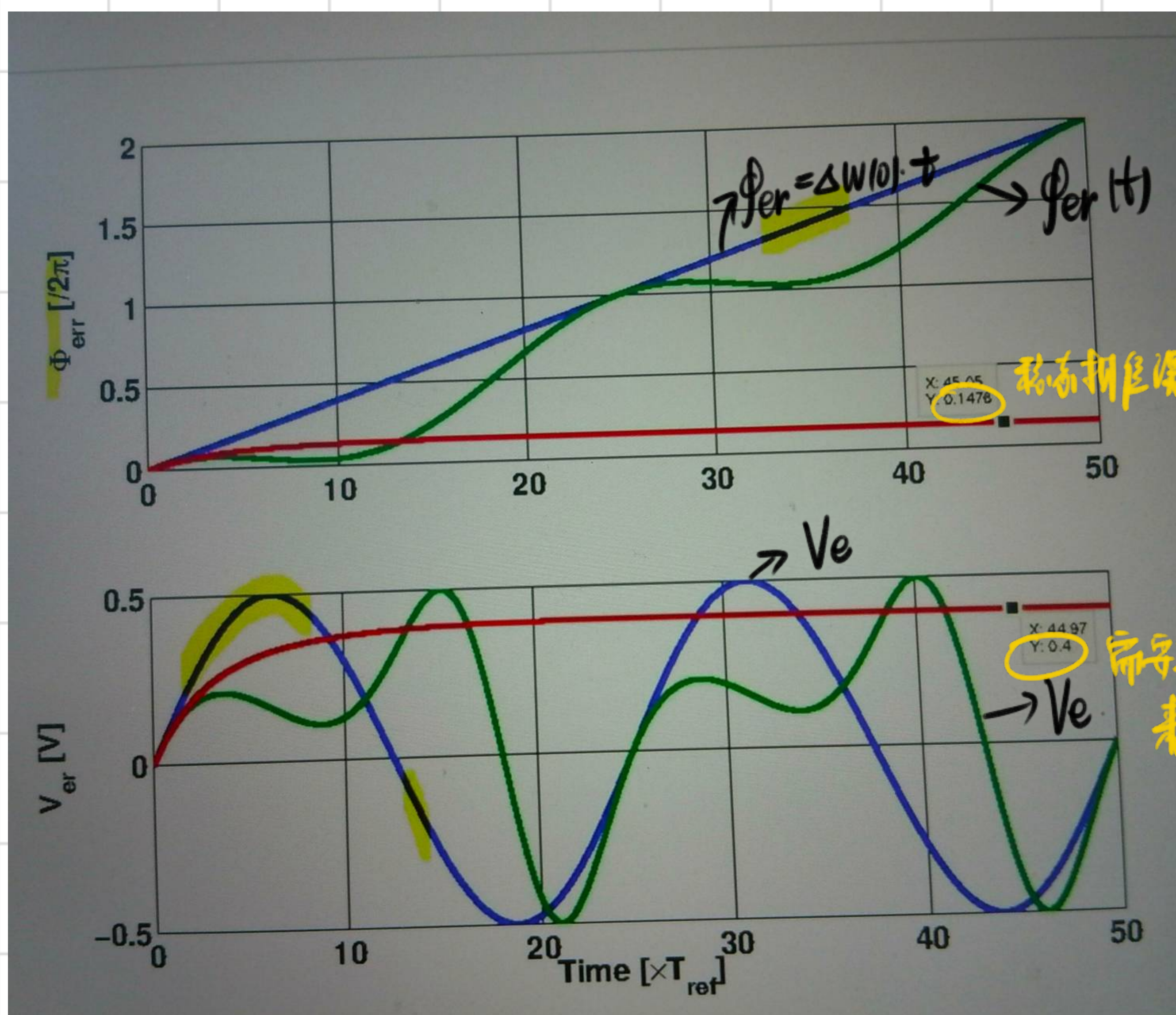
$\omega_{out} = \omega - \Delta\omega(0) - \frac{K_{vco}}{2} \sin(\Delta\omega(0) t)$

$\phi_{out}(t) = \int \omega_{out} dt$
 $= (\omega - \Delta\omega(0))t - \frac{K_{vco}}{2} \int \sin(\Delta\omega(0) t) dt$

$\phi_{err}(t) = \phi_{in}(t) - \phi_{out}(t)$

$= \omega t - \left[(\omega - \Delta\omega(0))t - \frac{K_{vco}}{2} \int \sin(\Delta\omega(0) t) dt \right]$

$= \Delta\omega(0) t + \frac{K_{vco}}{2} \int \sin(\Delta\omega(0) t) dt$



稳定状态:

$\omega_{out} = \omega_{in} = \omega - \Delta\omega + K_{vco} \cdot V_c$

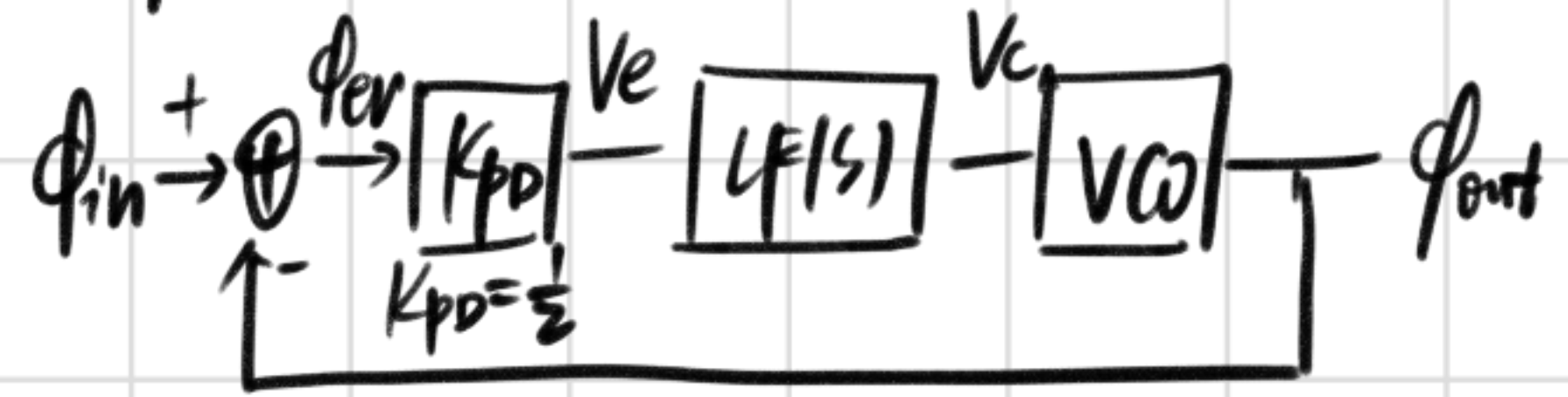
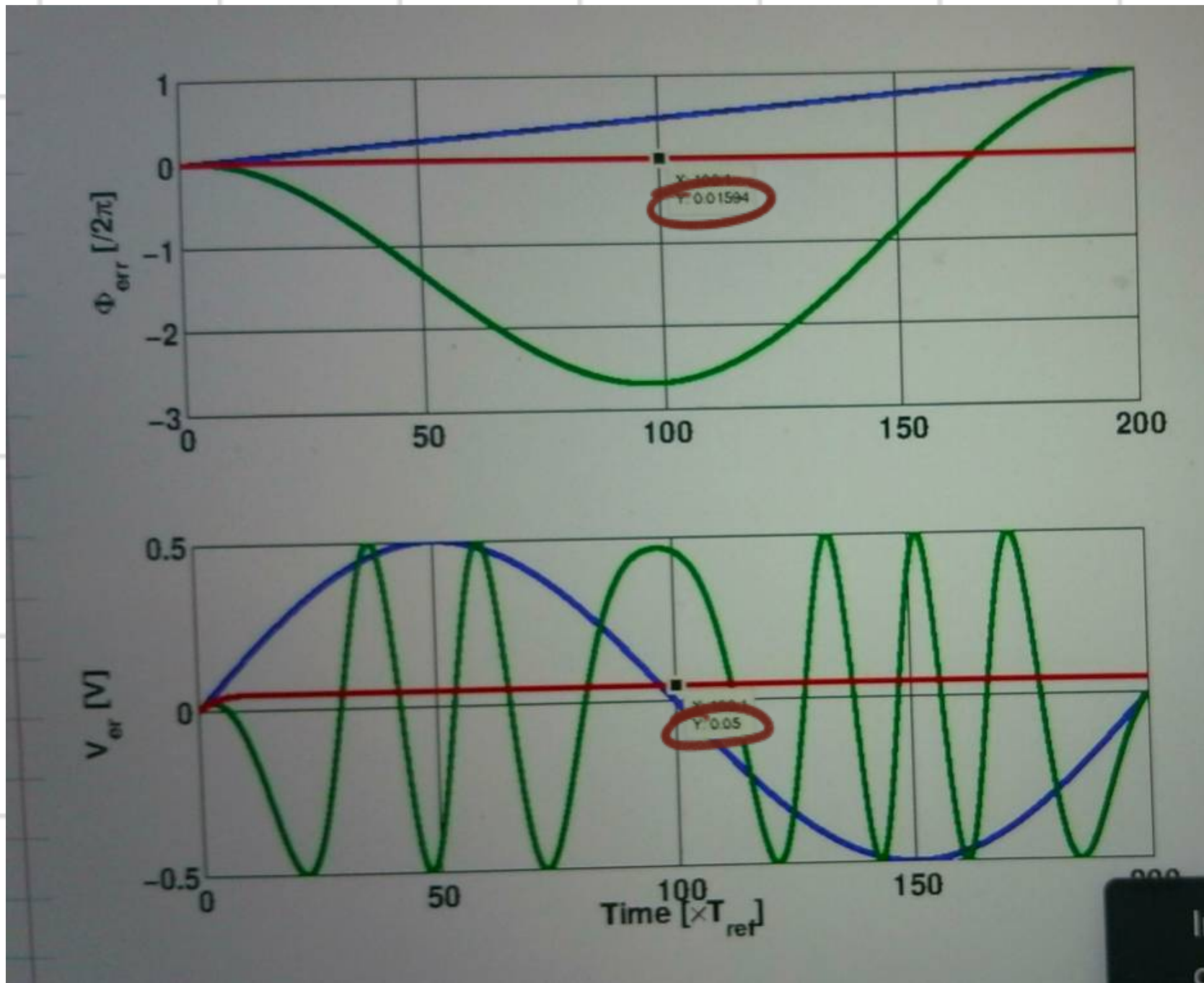
$\Rightarrow V_c = \frac{\Delta\omega}{K_{vco}}$

$V_c = \frac{1}{2} \sin(\phi_{err})$

$\phi_{err} = \sin^{-1}\left(\frac{2\Delta\omega}{K_{vco}}\right)$

需要输出一个恒定频率
来维持该频率差

Time Domain Versus Small Signal Analysis of a Simple PLL



at $t=0$, $W_{out} = W_{in} = W = 2\pi \times 45 \text{ M} \cdot \text{rad} \cdot \text{s}^{-1}$
 $\phi_{in} = \phi_{out}$, $\phi_{er}(t=0) = 0$, $V_c = 0$, $V_e = 0$.

at $t=0^+$, $\Delta W_{in} = 2\pi \times 5 \text{ M} \cdot \text{rad} \cdot \text{s}^{-1}$
 $\phi_{in} = \int W_{in} dt$ $\Rightarrow \phi_{in}(s) = \frac{W(s)}{s}$
 $\phi_{out} = \int \phi_{in} dt$ $\Rightarrow \phi_{out}(s) = \frac{\phi_{in}(s)}{s} = \frac{W(s)}{s^2}$



loop gain: $\frac{1}{2} \cdot \frac{1}{HsRC} \cdot \frac{K_{vco}}{s}$

closed gain: $\frac{\phi_{out}}{\phi_{in}} = \frac{\text{loop gain}}{1 + \text{loop gain}} = \frac{W_{out}}{W_{in}}$

$$\begin{cases} \Delta W_{in}(t) = \Delta W(0) u(t) \\ \Rightarrow \Delta W_{in}(s) = \frac{\Delta W(0)}{s} \\ \phi_{in}(s) = \frac{\Delta W(0)}{s^2} \end{cases}$$

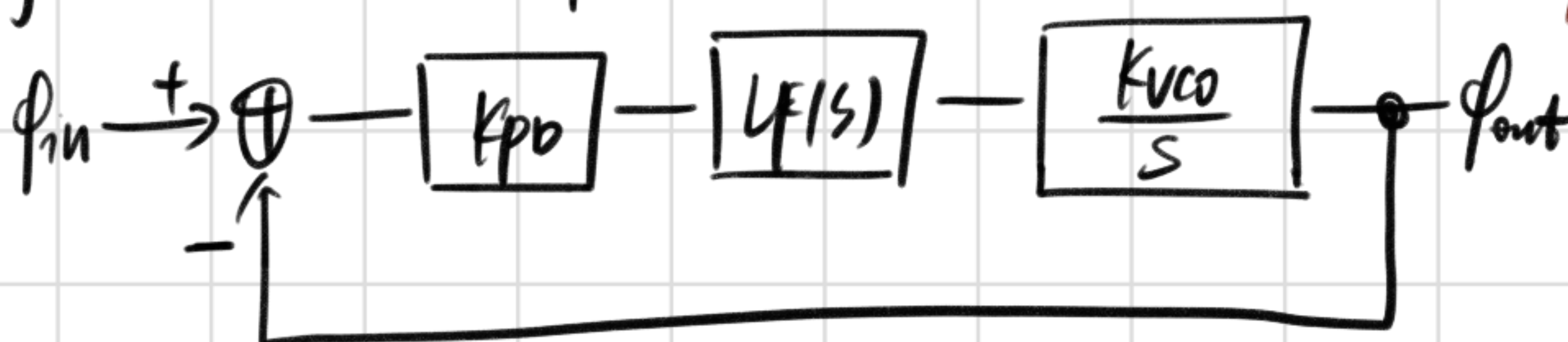
In steady state, $\lim_{t \rightarrow \infty} \phi_{er}(t) = \lim_{s \rightarrow 0} s \phi_{er}(s)$, where $\phi_{er}(s) = \phi_{in}(s) - \phi_{out}(s) = \frac{\phi_{in}(s)}{1 + \text{loop gain}}$

$$\begin{aligned} &= \lim_{s \rightarrow 0} s \cdot \frac{\phi_{in}}{1 + \frac{1}{2} \cdot \frac{1}{HsRC} \cdot \frac{K_{vco}}{s}} \\ &= \lim_{s \rightarrow 0} s \cdot \frac{\Delta W(0)}{s^2} \cdot \frac{1}{1 + \frac{1}{2} \cdot \frac{1}{HsRC} \cdot \frac{K_{vco}}{s}} \end{aligned}$$

$$= \frac{\Delta W(0)}{K_{vco}/2} = \frac{2\pi \times 5 \text{ M} \cdot \text{rad} \cdot \text{s}^{-1}}{2\pi \times 100 \text{ M} \cdot \text{rad} \cdot \text{s}^{-1} \cdot \text{V}^{-1} \times \text{V} \cdot \text{rad}^{-1}} = 0.1$$

Transfer function of loop filter
Important!!

Type and Order of PLL



Type I, order = 2.

Type: the number of integrators in the loop

Order: — of poles in the loop

$$LG(s) = K_{PD} \times LF(s) \times \frac{KV_{CO}}{s}, \text{ where } LF(s) = \frac{1}{1+sRC} \quad ①$$

$$= K_{PD} \times \frac{1}{1+sRC} \times \frac{KV_{CO}}{s}$$

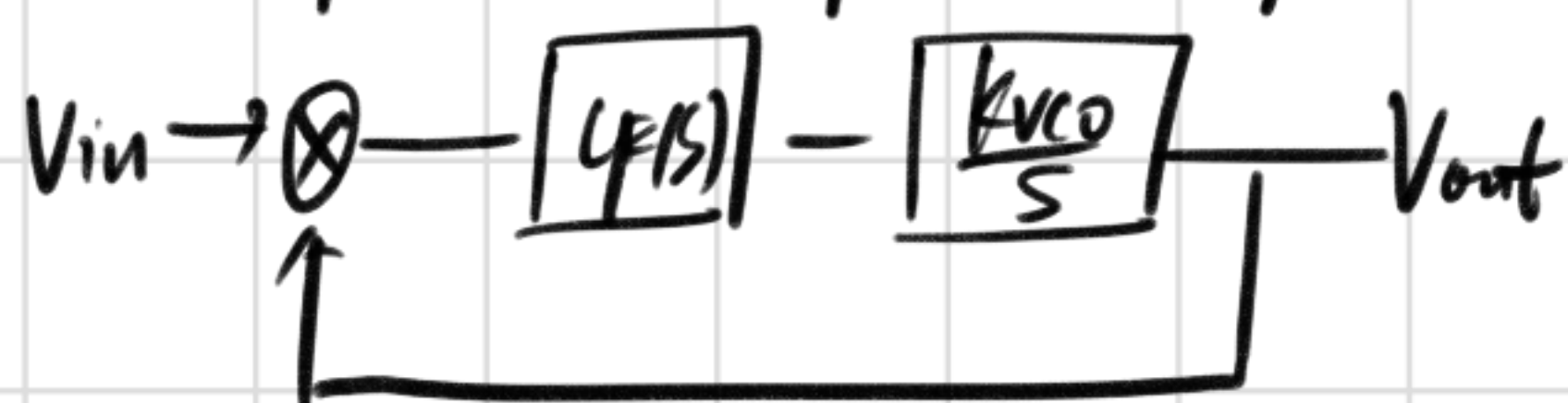
of integrator: 1

of pole: $s_1=0, s_2=-\frac{1}{RC} = -\frac{1}{RC}$

②: $LF(s) = \frac{1}{1+sT_1} \cdot \frac{1}{1+sT_2} \Rightarrow$ Type-I, order-3

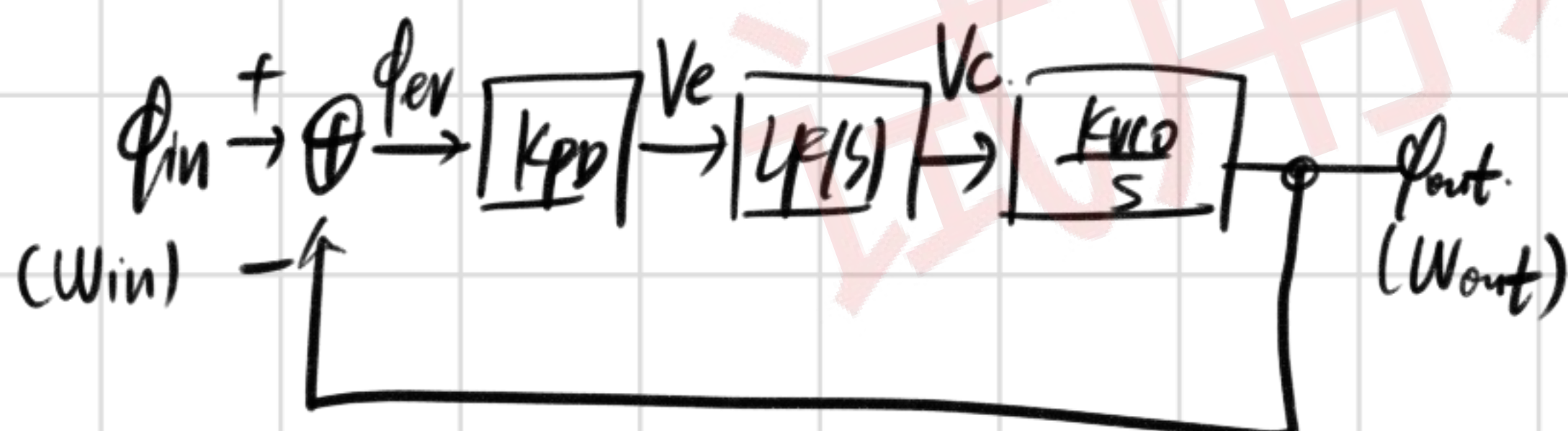
③: $LF(s) = \frac{1+sT_1}{sT_2} \Rightarrow$ Type-II, order 2.

Small Signal Analysis of Type PLLs for phase/frequency step, frequency ramp



$$V_{in} = \sin(\omega_{int} t)$$

$$V_{out} = \cos(\omega_{out} t)$$



at $t=0$, $\frac{d\phi_{err}}{dt} = 0$, $\omega_{in} = \omega_{out}$

at $t=0^+$, $\Delta\phi_{in} = \Delta\phi_{in} u(t)$

$$\lim_{t \rightarrow \infty} \phi_{err}(t) = \dots$$

Loop gain of PLL: $LG(s) = K_{PD} \cdot LF(s) \cdot \frac{KV_{CO}}{s}$

Input phase step, $\Delta\phi_{in} = \Delta\phi_{in}(0) u(t) \xrightarrow{f} \Delta\phi_{in}(s) = \frac{\Delta\phi_{in}(0)}{s}$

$$\begin{aligned} \lim_{t \rightarrow \infty} \phi_{err}(t) &= \lim_{s \rightarrow 0} s \phi_{err}(s) \\ &= \lim_{s \rightarrow 0} s \cdot \frac{1}{1+LG(s)} \phi_{in}(s) \\ &= \lim_{s \rightarrow 0} s \cdot \frac{1}{1 + \frac{K_{PD} K_{VCO}}{s} LF(s)} \cdot \frac{\Delta\phi_{in}(0)}{s} \\ &= \lim_{s \rightarrow 0} \frac{s}{s + K_{PD} K_{VCO} LF(s)} \\ &= 0 \end{aligned}$$

$$\begin{aligned} \phi_{err}(t) &= \phi_{in}(t) - \phi_{out}(t) \\ &= \phi_{in}(s) - \frac{1}{1+LG(s)} \phi_{in}(s) \\ &= \frac{\phi_{in}(s)}{1+LG(s)} \end{aligned}$$

$\lim_{t \rightarrow \infty} \phi_{err}(t) = 0$, for type-I PLL ★

Frequency step, $\Delta\omega_{in} = \Delta\omega_{in}(0) u(t)$

$$\Delta\phi_{in}(t) = \int_0^t \Delta\omega_{in} dt \xrightarrow{f} \Delta\phi_{in}(s) = \frac{\Delta\omega_{in}(0)}{s^2}$$

$$\begin{aligned} \lim_{t \rightarrow \infty} \phi_{err}(t) &= \lim_{s \rightarrow 0} s \phi_{err}(s) = \lim_{s \rightarrow 0} s \cdot \frac{\Delta\phi_{in}(s)}{1+LG(s)} \\ &= \lim_{s \rightarrow 0} s \cdot \frac{\Delta\omega_{in}(0)}{s^2} \times \frac{s}{s + K_{PD} K_{VCO} LF(s)} \\ &= \lim_{s \rightarrow 0} \frac{\Delta\omega_{in}(0)}{s + K_{PD} K_{VCO} LF(s)} = \frac{\Delta\omega_{in}(0)}{K_{PD} K_{VCO} LF(s)} \end{aligned}$$

Type-I PLL: $LF(0) = 1$, $LF(s) = \frac{1}{1+sRC}$

Type-II PLL: $LF(0) = \infty$, $LF(s) = \frac{1+sT_1}{sT_2}$

for freq step, $\phi_{err,\infty} = \lim_{t \rightarrow \infty} \phi_{err}(t)$

Type-I: $\phi_{err,\infty} = \frac{\Delta\omega_{in}(0)}{K_{PD} \cdot K_{VCO}}$ — Type-I 对 $\Delta\omega_{in}(0)$ 的范围有限制

Type-II: $\phi_{err,\infty} = 0$

Frequency Ramp

$$\Delta W_{in}(t) = \Delta W_{in} t = \int \Delta W_{in}(0) dt \Leftrightarrow \Delta W_{in}(s) = \frac{\Delta W_{in}(0)}{s^2}$$

$$\Delta \Phi_{in}(s) = \frac{\Delta W_{in}(0)}{s^3}$$

$$\begin{aligned} \Phi_{err,s} &= \lim_{t \rightarrow \infty} \Phi_{err}(t) = \lim_{s \rightarrow 0} s \cdot \Phi_{err}(s) = \lim_{s \rightarrow 0} s \cdot \frac{\Delta W_{in}(0)}{s^3} \cdot \frac{s}{s + K_p K_{vco} Lf(s)} \\ &= \lim_{s \rightarrow 0} \frac{\Delta W_{in}(0)}{s^2 + s K_p K_{vco} Lf(s)} \\ &= \lim_{s \rightarrow 0} \frac{\Delta W_{in}(0)}{s K_p K_{vco} Lf(s)} \end{aligned}$$

$$\text{Type I: } Lf = \frac{1}{1 + s\tau_1},$$

$$\Phi_{err,s} = \dots \rightarrow \infty, \text{ pll not locked}$$

$$\text{Type II: } Lf = \frac{1 + s\tau_1}{s\tau_2}$$

$$\Phi_{err,s} = \lim_{s \rightarrow 0} \frac{\Delta W_{in}(0)}{s K_p K_{vco} \frac{1 + s\tau_1}{s\tau_2}} = \frac{\Delta W_{in}(0)}{K_p K_{vco} \tau_2}$$

$$\text{Type III: } Lf(s) = (1 + s\tau_1)(1 + s\tau_2) \cdot \frac{1}{s^2 \tau_{i1} \tau_{i2}}$$

$$\Phi_{err,s} = \dots \rightarrow 0$$

$\Phi_{err,s}$	Phase Step	Frequency Step	Frequency Ramp
Type I	0	$\frac{\Delta W}{K_p K_{vco}}$	∞
Type II	0	0	$\frac{\Delta W}{K_p K_{vco} \tau_2}$
Type III	0	0	0

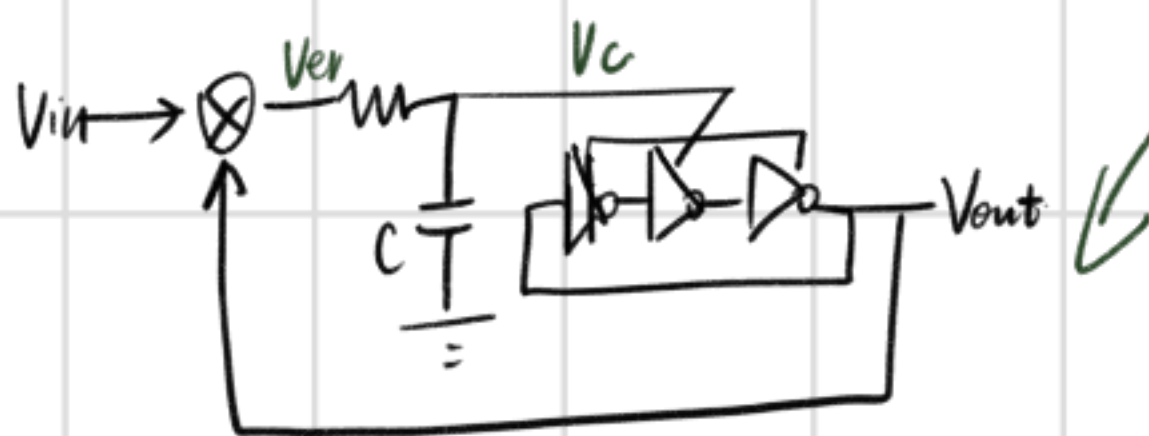
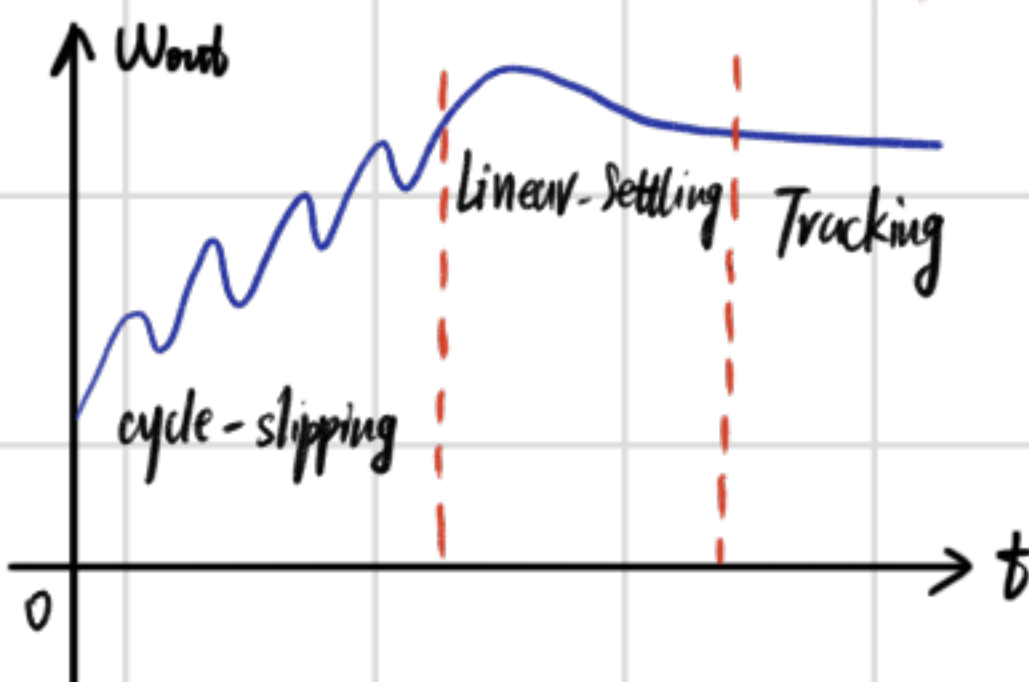
Frequency Acquisition Range for PLL

Acquisition of clock in PLLs:



$$W_{out} = W_{fr} + K_{vco} \cdot V_c$$

at $t=0$, $W_{in} = W_0$, $W_{fr} \neq W_0$ for $V_c = 0$.



$$V_{in} = \sin(W_{in} t)$$

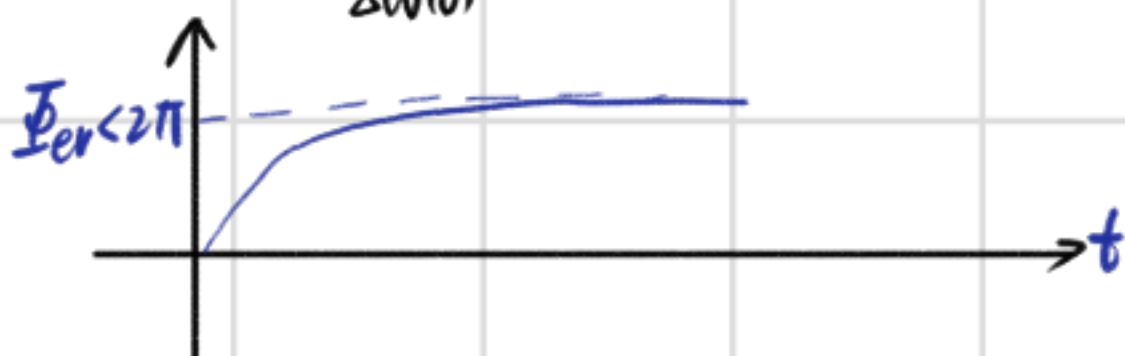
$$V_{out} = \cos(W_{out} t)$$

at $t=0$, $W_{out} = W_{fr}$, $V_c = 0$, $V_{er} = 0$, $W_{in} \neq W_{out}$

$$\Phi_{err} = \Phi_{in}(t) - \Phi_{out}(t)$$

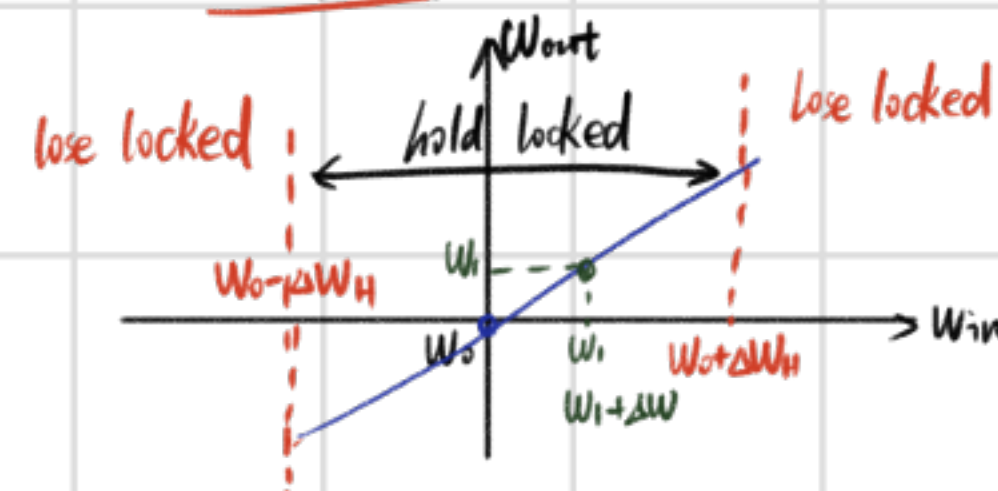
$$= \int W_{in} dt - \int_0^t (W_{fr} + K_{vco} V_c(\tau)) d\tau$$

$$\Rightarrow \Phi_{err} = \frac{(W_{in} - W_{fr})t}{\Delta W(0)} - \int_0^t K_{vco} V_c(\tau) d\tau$$



Acquisition Range:

1. Holding Range (ΔW_H): The range of frequency that the pll can stay in lock.

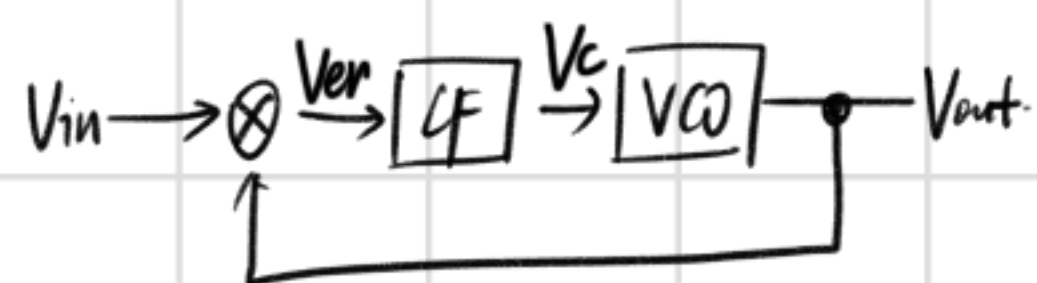


2. Lock-in Range (ΔW_L): The range of frequency over which the PLL can acquire lock with cycle slipping or without phase error exceeding 2π .

3. Pull-in-Range (ΔW_P): The range of frequency over which the PLL can acquire lock with or without cycle slipping is called the holding range.

$$\text{一般的: } \Delta W_L \leq \Delta W_P \leq \Delta W_H$$

Frequency Acquisition in Type-I PLLs



$$V_{in} = \sin(\omega_{in} \cdot t)$$

$$V_{out} = \cos(\omega_{out} \cdot t)$$

$$V_{er} = \frac{1}{2} [\sin(\omega_{in} + \omega_{out} t) + \sin(\omega_{in} - \omega_{out} t)]$$

$$\text{at } t=0, \omega_{in} = \omega, \omega_{out} = \omega_{fr} + K_{vco} \cdot V_c$$

$$V_c = 0, \omega_{out} = \omega_{fr}$$

$$\Delta\omega = \omega_{in} - \omega_{out} = (\omega - \omega_{fr}) - K_{vco} \cdot V_c$$

$$\Phi_{er} = \int_0^t \Delta\omega dt = \int_0^t (\Delta\omega(0) - K_{vco} V_c) dt$$

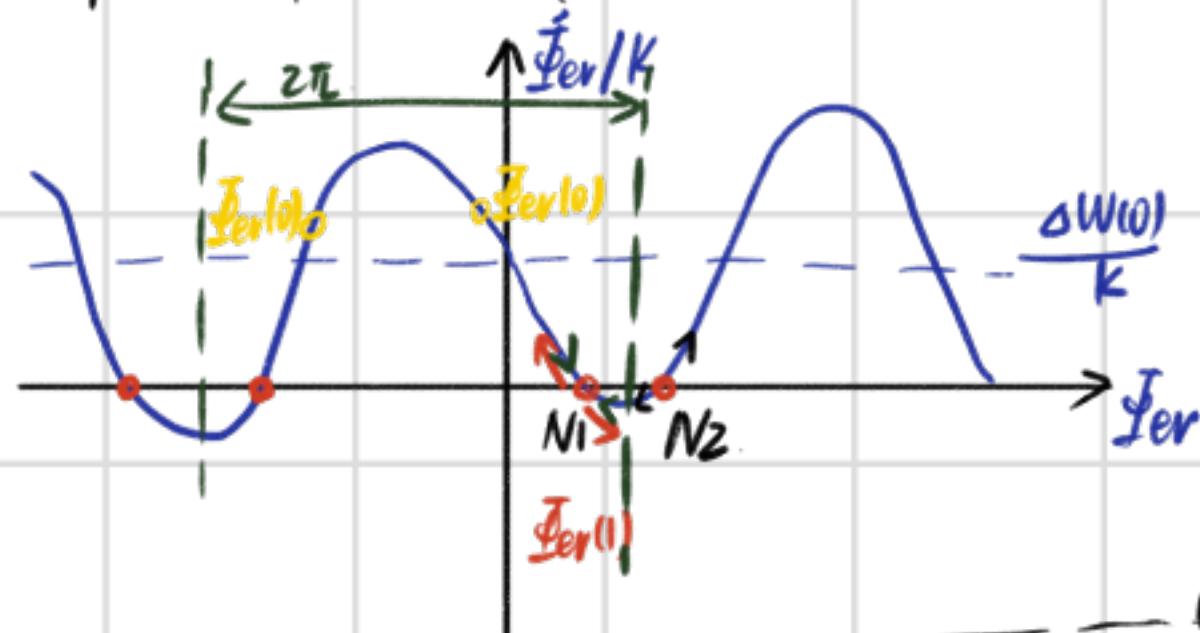
When PLL is locked, $\frac{d\Phi_{er}}{dt} = 0$

$$\frac{d\Phi_{er}}{dt} = \dot{\Phi}_{er} = \Delta\omega(0) - K_{vco} V_c$$

$$\text{If } V_c = \frac{1}{K_{vco}} \sin(\Phi_{er})$$

$$\dot{\Phi}_{er} = \Delta\omega(0) - K_{vco} \cdot \frac{1}{K_{vco}} \sin(\Phi_{er})$$

$$\frac{\dot{\Phi}_{er}}{K} = \frac{\Delta\omega(0)}{K} - \sin(\Phi_{er}) = 0$$



① at N_1 , $\Phi_{er} = \Phi_{er}(1) + \Delta\phi \xrightarrow{\dot{\Phi}_{er} < 0} \Phi_{er} \downarrow$
if $\Phi_{er} = \Phi_{er}(1) - \Delta\phi \xrightarrow{\dot{\Phi}_{er} > 0} \Phi_{er} \uparrow$
 $\Rightarrow N_1$ is a stable solution.

② at N_2 , $\dot{\Phi}_{er} = 0$
if $\Phi_{er} = \Phi_{er}(2) + \Delta\phi \xrightarrow{\dot{\Phi}_{er} > 0} \Phi_{er} \uparrow$
if $\Phi_{er} = \Phi_{er}(2) - \Delta\phi \xrightarrow{\dot{\Phi}_{er} < 0} \Phi_{er} \downarrow$
 $\Rightarrow N_2$ is not a stable solution!

only has solutions when $\frac{\dot{\Phi}_{er}}{K} \leq 0 \Rightarrow \frac{\Delta\omega(0)}{K} \leq \sin(\Phi_{er})$

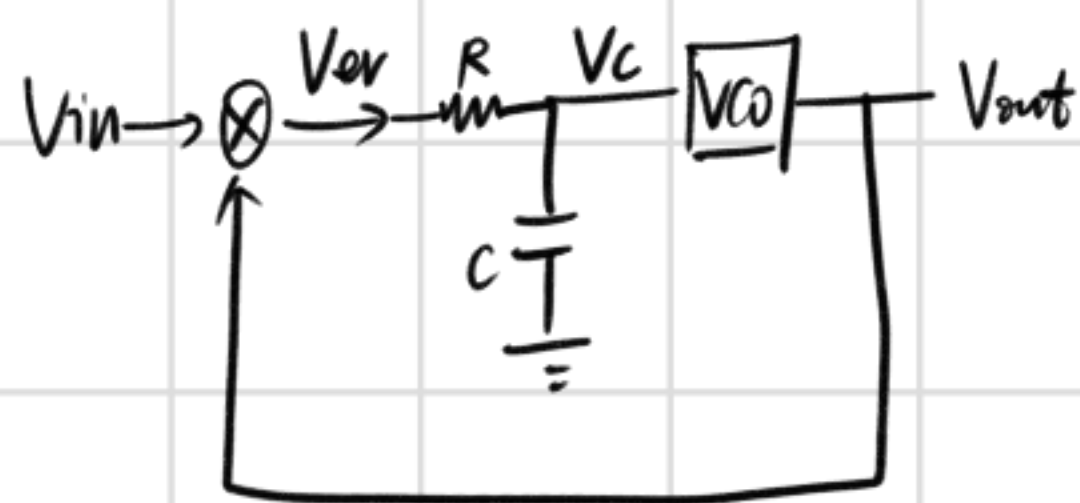
$$\Rightarrow \frac{\Delta\omega(0)}{K} \leq 1 \quad \Delta\omega(0) \text{ 有上限}$$

$\frac{\dot{\Phi}_{er}}{K} = 0$ has two solutions: N_1, N_2

If $\frac{\Delta\omega(0)}{K} \leq 1$, there stays a solution for $-\pi + \Phi_{os} \leq \Phi_{er} \leq \pi + \Phi_{er}$

$\Leftrightarrow$ lock-in range: $\Delta\omega_L = K = K_{p0} K_{vco}$

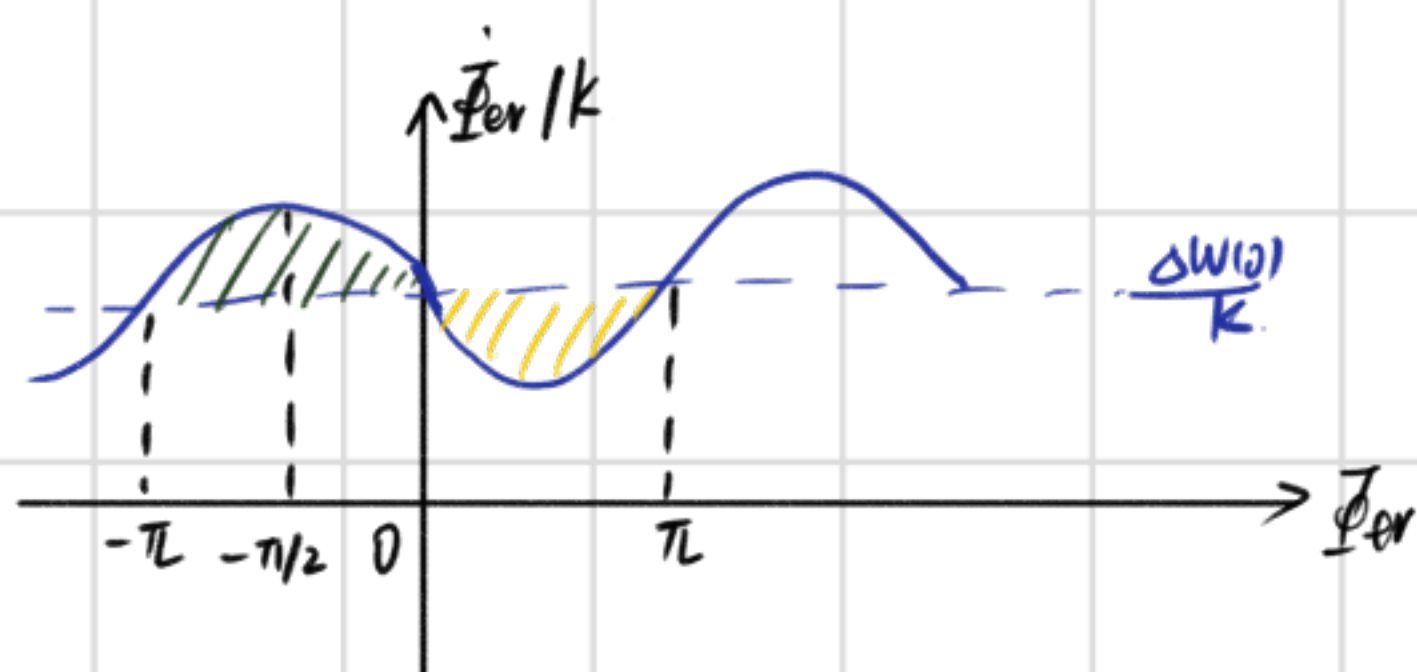
Frequency Acquisition limits in Type-I PLL



$$\frac{\dot{\Phi}_{er}}{K} = \frac{\Delta\omega(0)}{K} - \sin(\Phi_{er})$$

$$\text{at } t=0, \omega_{in} = \omega, \omega_{out} = \omega_{fr}, V_c = 0$$

$$\Delta\omega(0) = \omega - \omega_{fr}$$



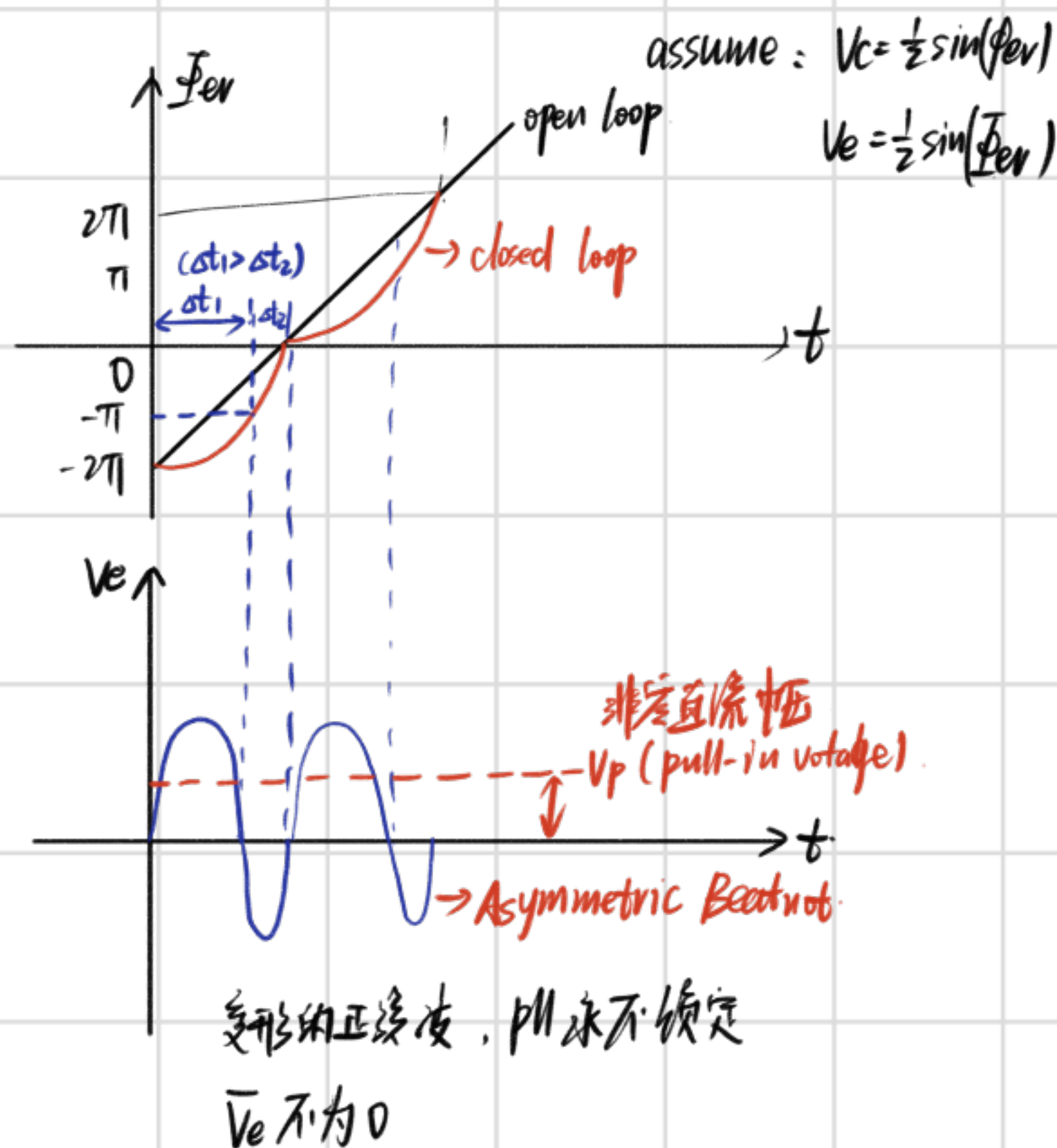
$$\text{at } \Phi_{er} = -\frac{\pi}{2}, \dot{\Phi}_{er} = \Delta\omega(0) + K$$

$$\text{at } \Phi_{er} = \frac{\pi}{2}, \dot{\Phi}_{er} = \Delta\omega(0) - K$$

$$V_{in} \rightarrow \otimes \xrightarrow{V_{er}} V_c \xrightarrow{V_{out}} V_{out}$$

$$\Phi_{er} = \Delta\omega(0)t$$

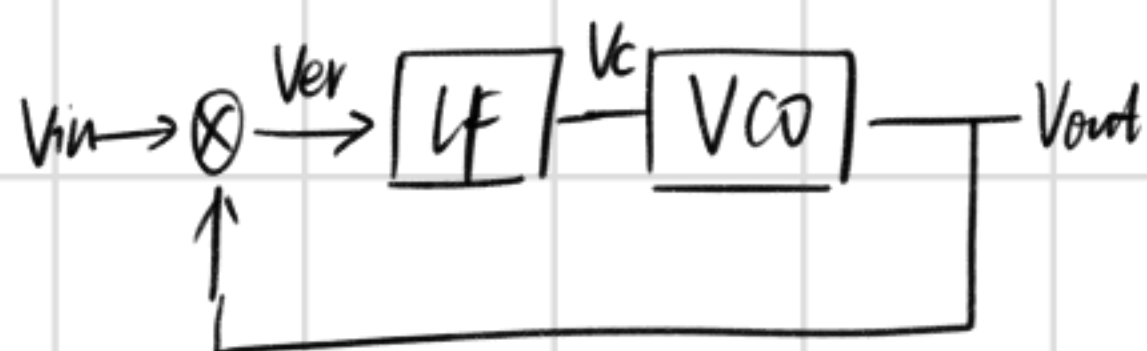
$$\Delta\omega(0)t = 2\pi, t = \frac{2\pi}{\Delta\omega(0)}$$



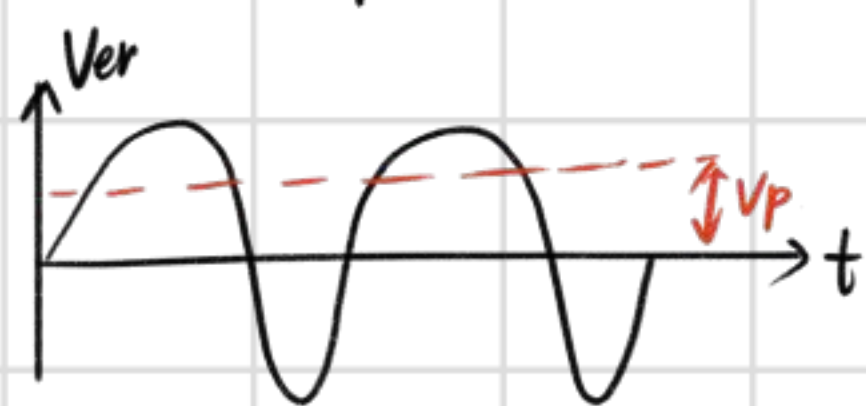
In open loop, $\text{avg}(V_e) = 0$

In close loop, $\text{avg}(V_e) \neq 0$
even when $\frac{\Delta\omega(0)}{K} > 1$

Frequency Acquisition in Type-II PLLs



for $\Delta W(0) > K_{pd} K_{vco}$

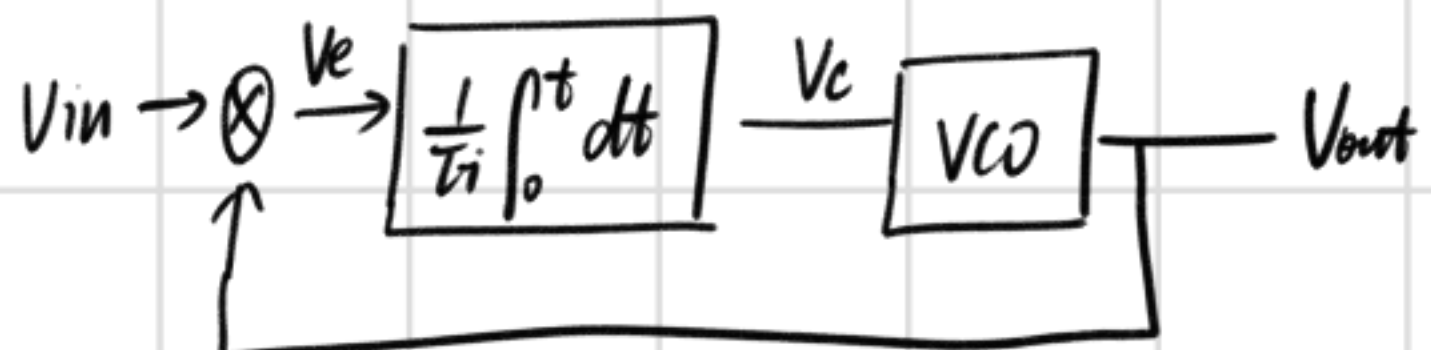


$$W_{in} = W_{out}$$

$$W = W_{fr} + K_{vco} \cdot V_c$$

In steady state, $V_c = \frac{W - W_{fr}}{K_{vco}} = \frac{\Delta W(0)}{K_{vco}}$

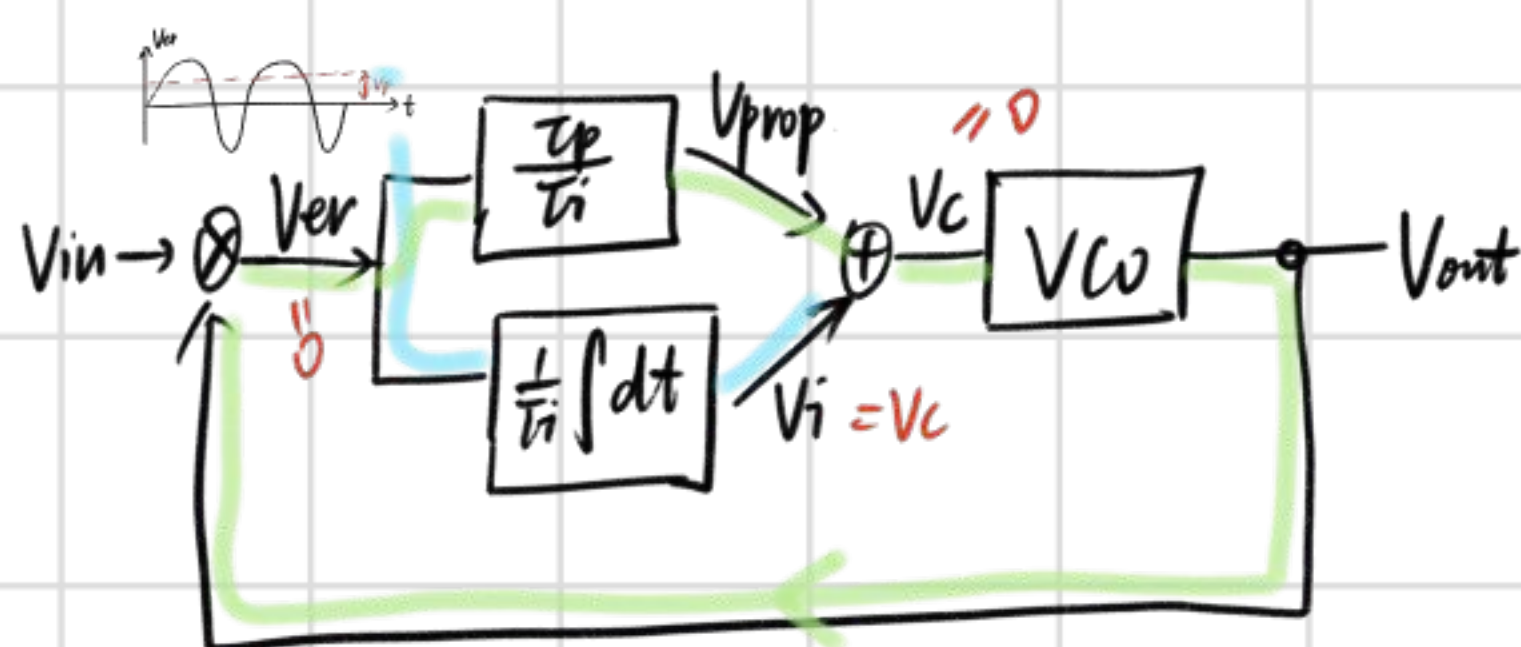
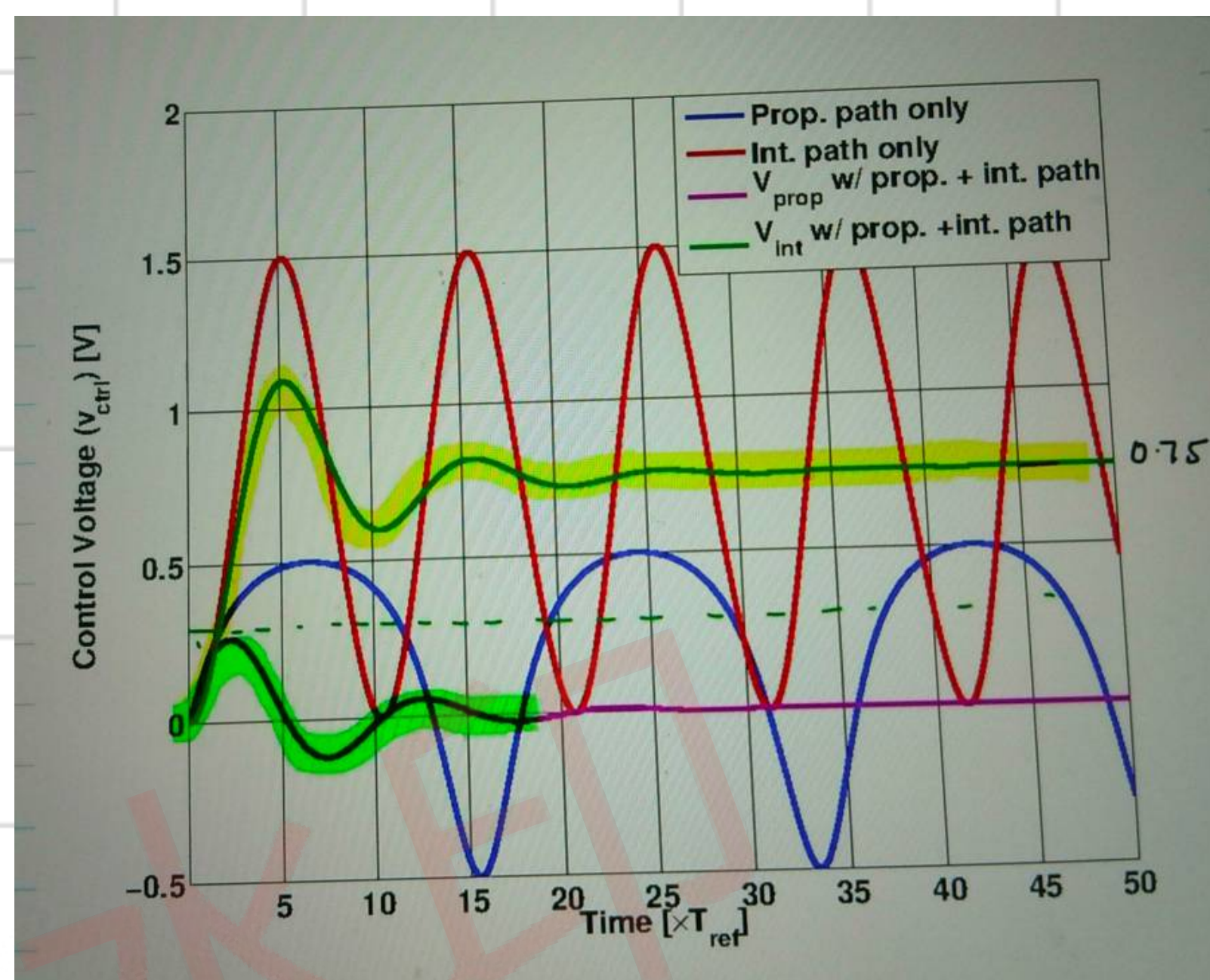
- Integrator gives infinite dc gain:



at $t=0$, $\Delta W = W_{in} - W_{fr} = \Delta W(0)$, $V_e = 0$, $V_c = 0$.

$$V_e = \frac{1}{2} \sin(\Delta W(0)t) + \sin(\underbrace{W_{in} + W_{out}}_{\text{neglect}} t)$$

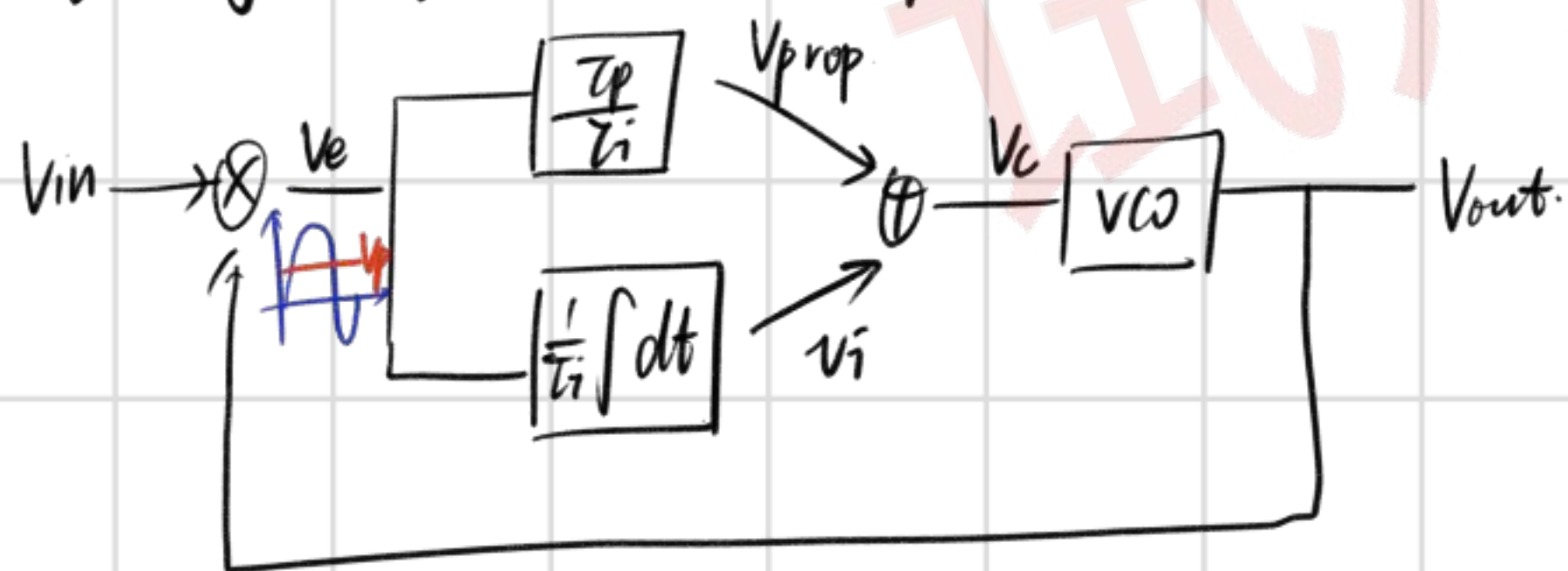
$$\int V_e dt = \frac{1}{2} \int_0^t \sin(\Delta W t) dt \rightarrow 0$$



$$V_c = \frac{T_p}{s_i} V_e + \frac{1}{s_i} \int V_e dt$$

$$W_{out} = [K_{vco} (\frac{T_p}{s_i}) V_e + K_{vco} \int_0^t V_e dt] + W_{fr}$$

Frequency Acquisition Ranges in Type-II PLLs with ideal & non-ideal Integrator



$$V_{in} = \sin(W_{in} t)$$

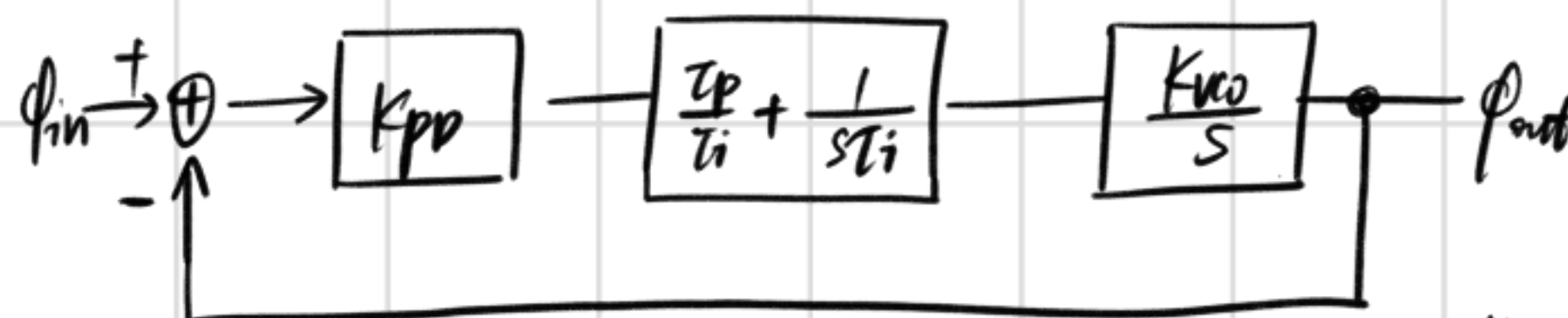
$$V_{out} = \cos(W_{out} t)$$

at $t=0$, $W_{in} = W$, $W_{out} = W_{fr}$

$$W_{in} - W_{out} = W - W_{fr} = \Delta W$$

In steady state ($t \rightarrow \infty$):

$$\begin{cases} V_c = \frac{\Delta W(0)}{K_{vco}} = V_i \\ V_e = 0 \end{cases}$$

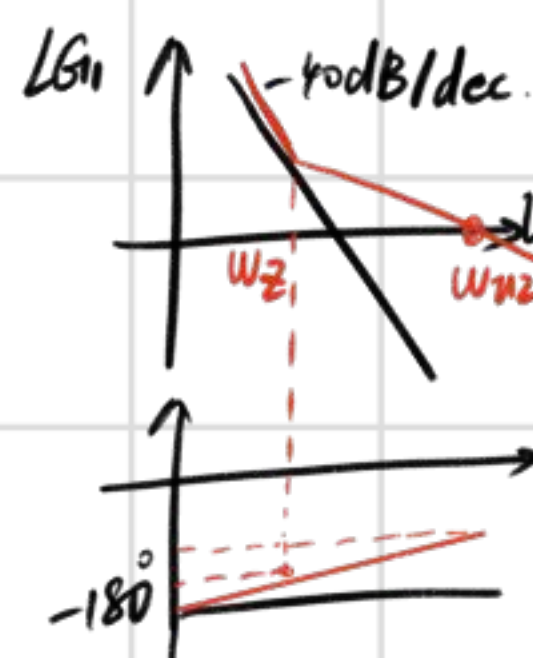


small signal

$$L_G(s) = K_{pd} \left(\frac{H s T_p}{s T_i} \right) \frac{K_{vco}}{s} = \frac{K_{pd} K_{vco}}{s^2 T_i} (H s T_p)$$

In case only has integrator, $T_p = 0$

$$L_{G1} = K_{pd} \cdot \frac{1}{s T_i} \cdot \frac{K_{vco}}{s} \quad \text{un-stable!}$$

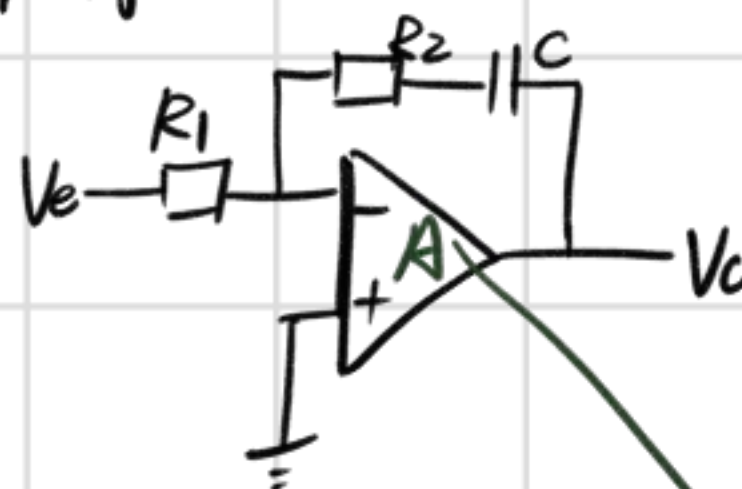


for L_{G1} , $PM = 0$

for L_G , $W_2 = \frac{1}{T_p}$

$$PM = -180^\circ + \tan^{-1} \left(\frac{W_{u2}}{W_2} \right) - (-180^\circ) = \arctan \left(\frac{W_{u2}}{W_2} \right) \quad \text{stable!!}$$

$$\frac{T_p}{s_i} + \frac{1}{s_i} \int dt \xrightarrow{s} \frac{T_p}{s_i} + \frac{1}{s T_i}$$



$$\text{where } T_p = C R_2, \quad T_i = C R_1$$

- Assuming OPAMP Ideal

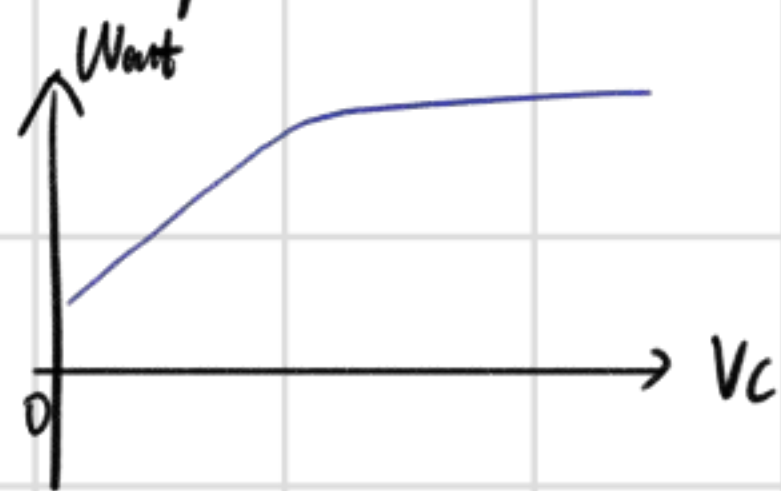
$$\frac{V_c(s)}{V_e(s)} = - \left(\frac{H s C R_2}{s C R_1} \right)$$

$$\approx \frac{V_c(s)}{V_e(s)} = \frac{H s T_p}{s T_i}$$

$$\frac{V_c(s)}{V_e(s)} = - \frac{A(H s T_p)}{H s T_p + (H A) s T_i} \quad A \rightarrow \infty \quad \text{non-ideal!}$$

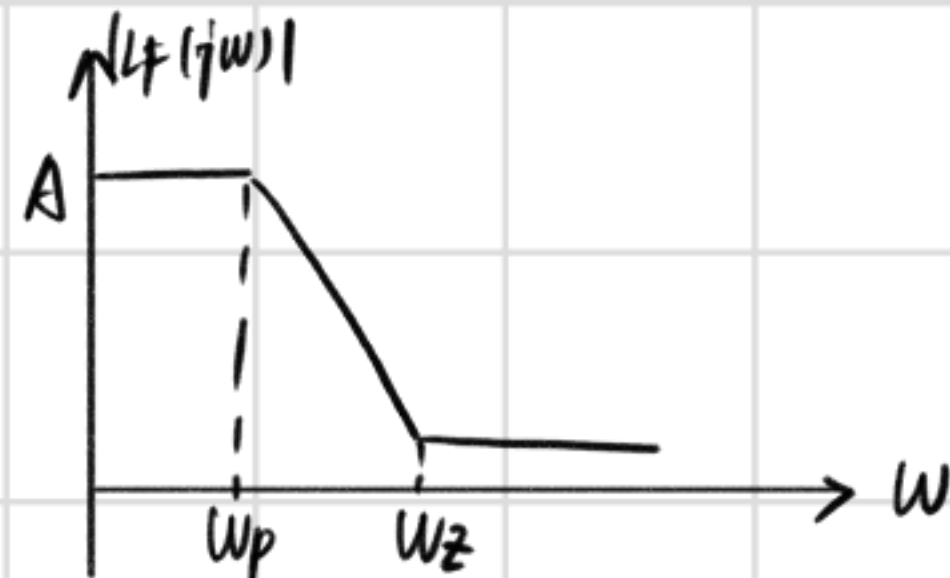
How large is the $\Delta W(0)$?

No limit!! ($\Delta W(0) \rightarrow \infty$)



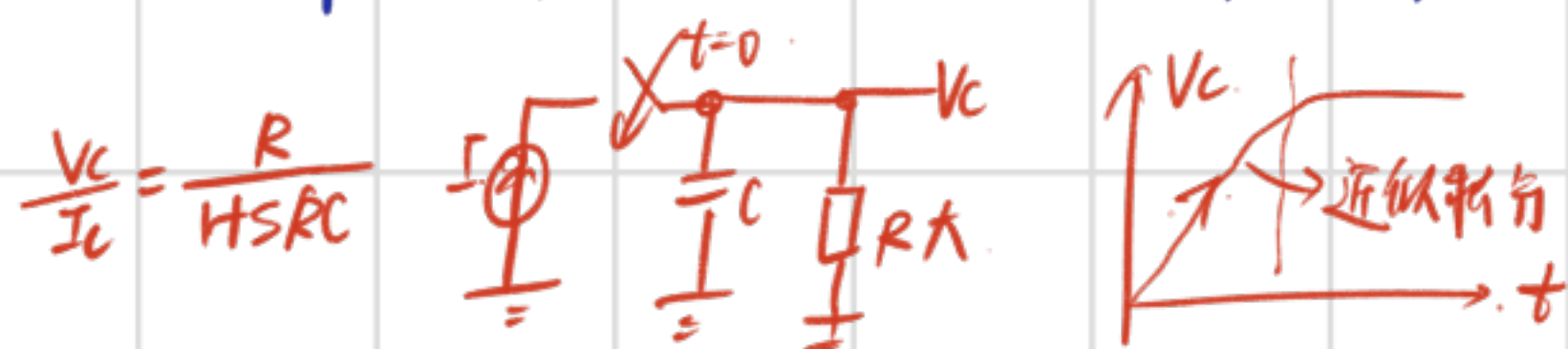
$$\begin{cases} \Delta W_p \rightarrow \infty \quad (\text{Pull-in}) \\ \Delta W_H \rightarrow \infty \quad (\text{Hold-in}) \\ \Delta W_L = K_{pd} \left(\frac{T_p}{s_i} \right) K_{vco} \quad (\text{Lock in range}) \end{cases}$$

$$\frac{V_c}{V_e}(s) = L_F(s)$$



$$\lim_{s \rightarrow \infty} L_F(s) = -\frac{A(\frac{1}{s} + \tau_p)}{\frac{1}{s} + \tau_p + (HA)\tau_i} = -\frac{A\tau_p}{\tau_p + (HA)\tau_i} \xrightarrow{A \rightarrow \infty} \frac{\tau_p}{\tau_i}$$

$$\Rightarrow L_F(s) = \underbrace{\frac{A\tau_p}{\tau_p + (HA)\tau_i}}_{\text{prop path gain}} + \underbrace{\frac{A(HA)\tau_i}{\tau_p + (HA)\tau_i}}_{\text{nonideal integrator path}} \cdot \frac{1}{(1+s\tau_p + (HA)s\tau_i)}$$



for loop filter LF:

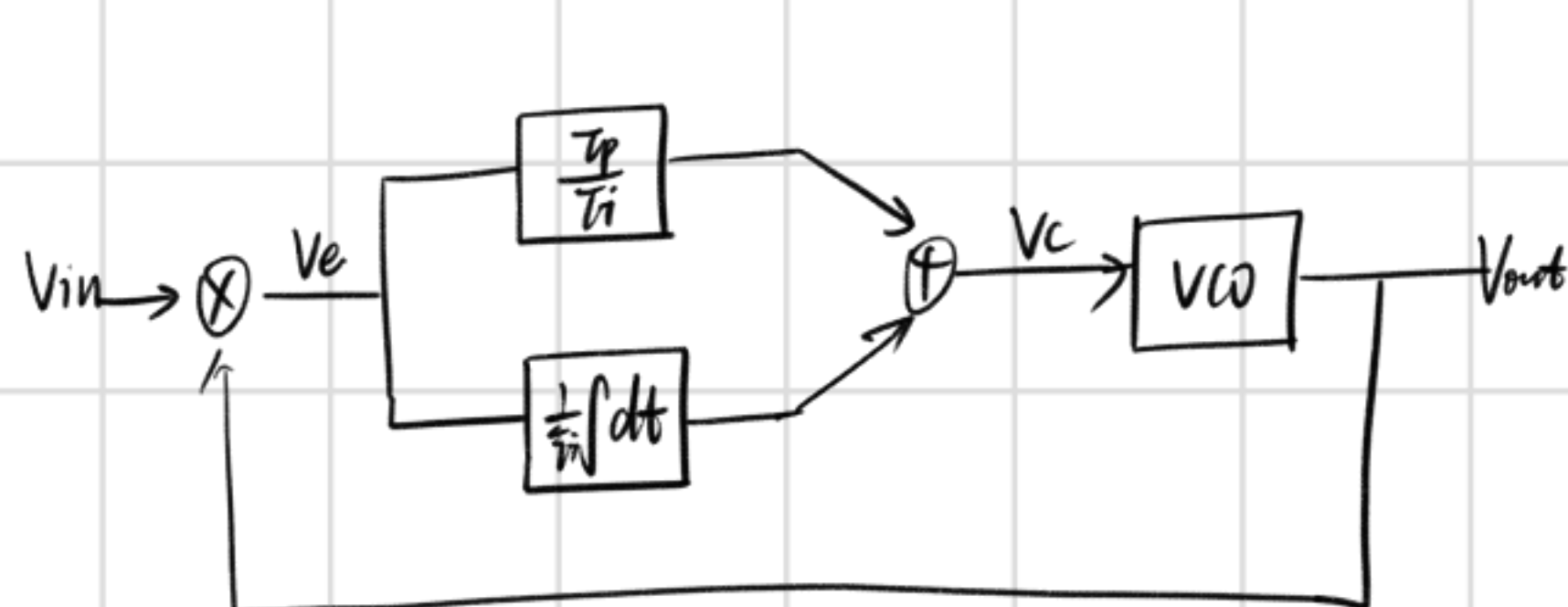
$$\Delta\omega_p = K_{PD}K_{VCO} \sqrt{2L_F(0)L_F(\infty)}$$

$$\Delta\omega_H = K_{PD}K_{VCO} L_F(0)$$

— Mixer is PD, loop filter, VCO has no frequency limits, input and output signals are sinusoids.

— Approximate limit.

Frequency Domain Insight in Frequency Acquisition for Type-II PLLs



at $t=0$, $\omega_{in} = \omega$, $\omega_{out} = \omega_{fr} + K_{VCO}V_c = \omega_{fr}$
 $\Delta\omega(0) = \omega - \omega_{fr}$

$$V_e = \frac{1}{s} [\sin(\omega_{in}t + \omega_{out}t) + \sin(\Delta\omega(0)t)]$$

$$V_c = K_{PD} \sin(\Delta\omega t)$$

$$V_{out} = \cos(\omega_{fr}t + K_{VCO} \int_0^t V_c(\tau) d\tau)$$

$$= \cos(\omega_{fr}t + K_{VCO} \int_0^t K_{PD} \sin(\Delta\omega\tau) d\tau)$$

$$= \cos(\omega_{fr}t + \frac{K_{VCO}K_{PD}}{\Delta\omega} (-\cos(\Delta\omega t)))$$

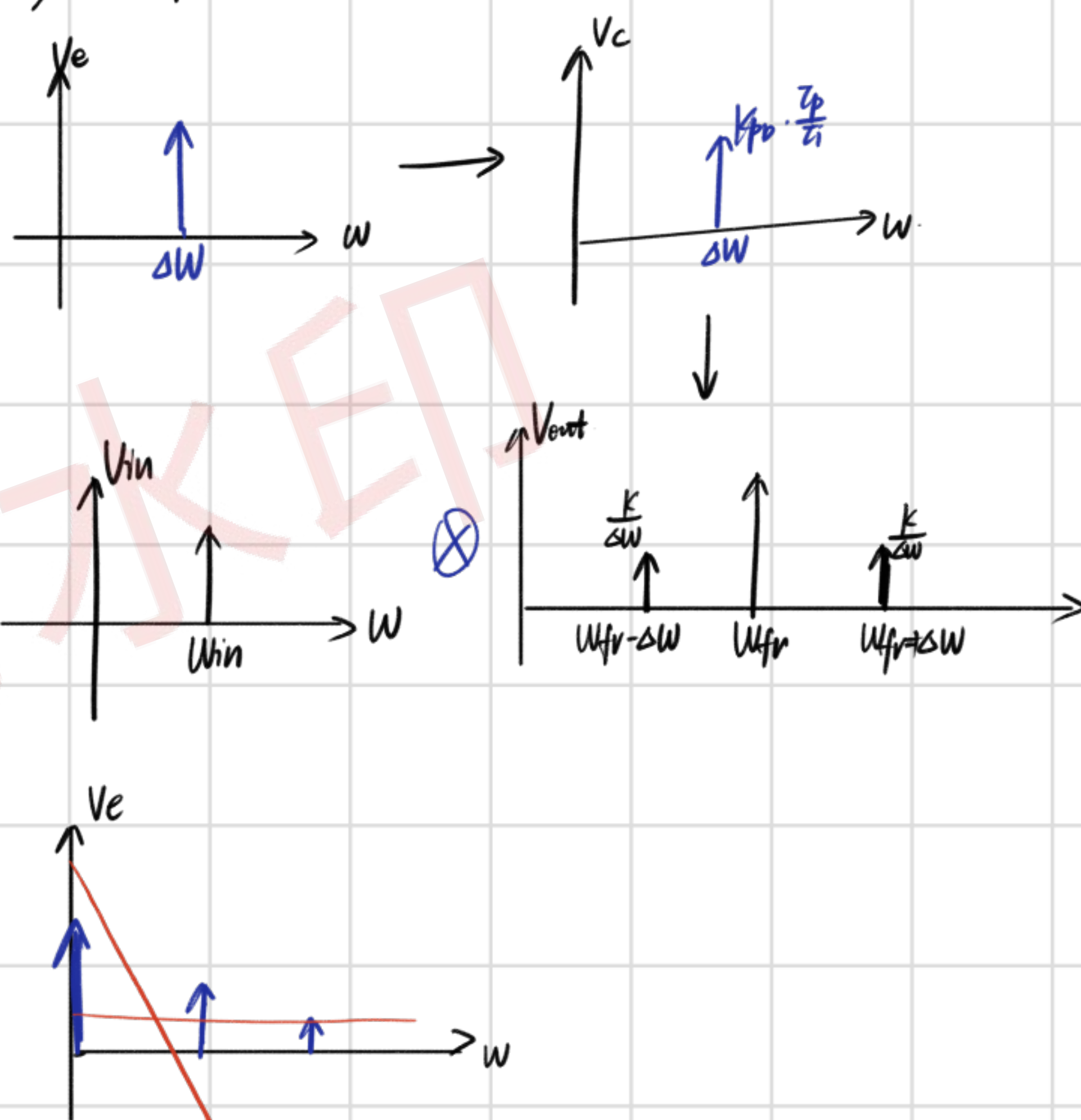
$$= \cos(\omega_{fr}t - \frac{K}{\Delta\omega} \cos(\Delta\omega t))$$

$$= \cos(\omega_{fr}t) \cdot \cos(\frac{K}{\Delta\omega} \cos(\Delta\omega t)) + \sin(\omega_{fr}t) \sin(\frac{K}{\Delta\omega} \cos(\Delta\omega t))$$

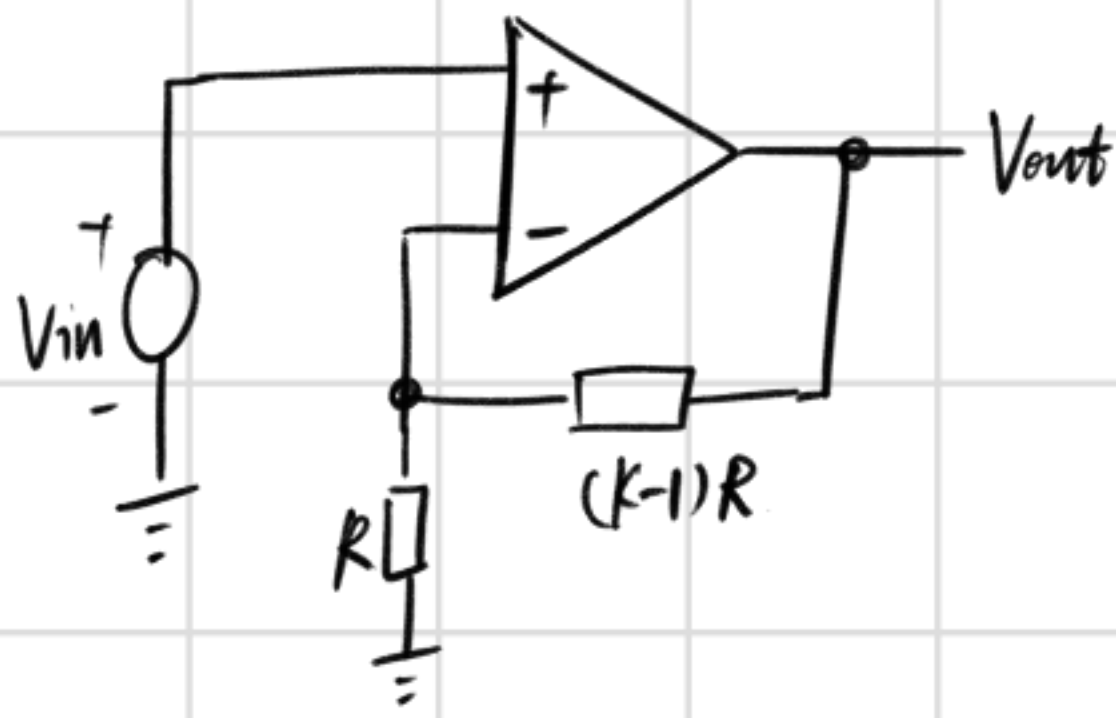
Assume $\frac{K}{\Delta\omega} \cos(\Delta\omega t)$ is very small

$$V_{out} = \cos(\omega_{fr}t) + \sin(\omega_{fr}t) \cdot \frac{K}{\Delta\omega} \cos(\Delta\omega t)$$

$$= \cos(\omega_{fr}t) + \frac{K}{2\Delta\omega} [\sin(\omega_{fr} + \Delta\omega)t + \sin(\omega_{fr} - \Delta\omega)t]$$



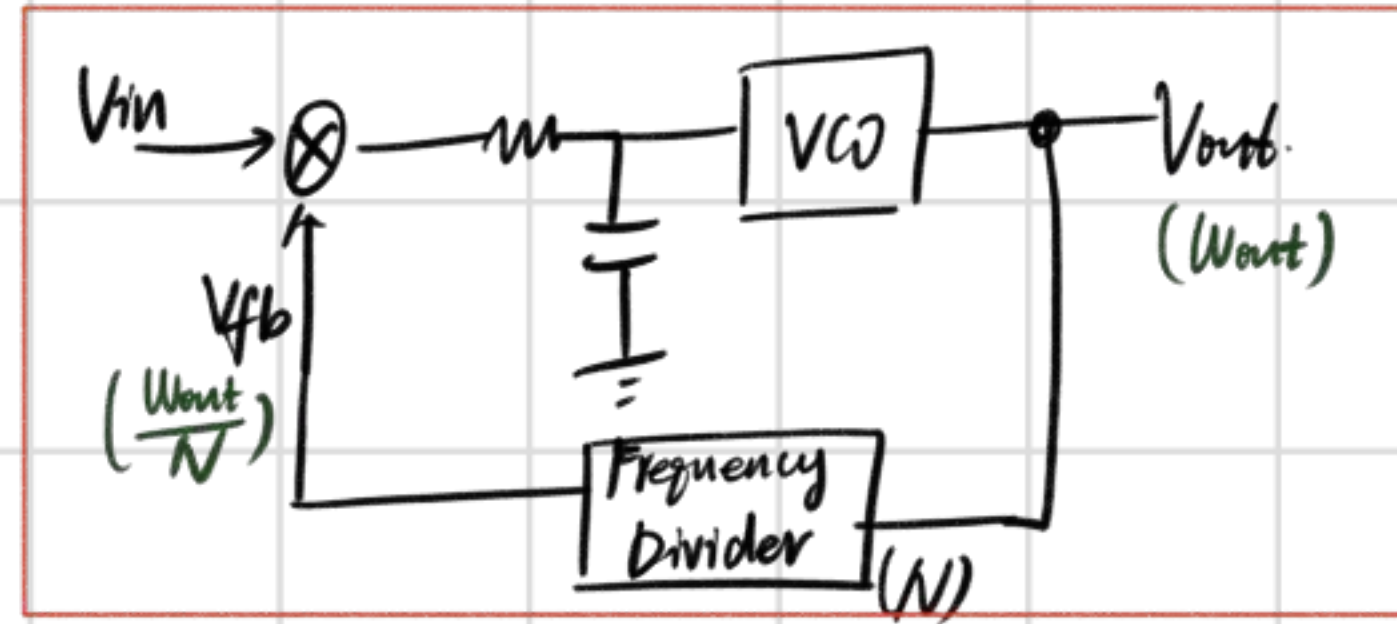
Clock Multiplier



$$\frac{V_{out}}{V_{in}} = K$$

$$V_{fb} = \frac{V_o}{K} = V_{in}$$

We're comparing $\frac{V_{out}}{K} = V_{fb}$ with V_{in}



$$V_{in} = \sin(W_{in}t)$$

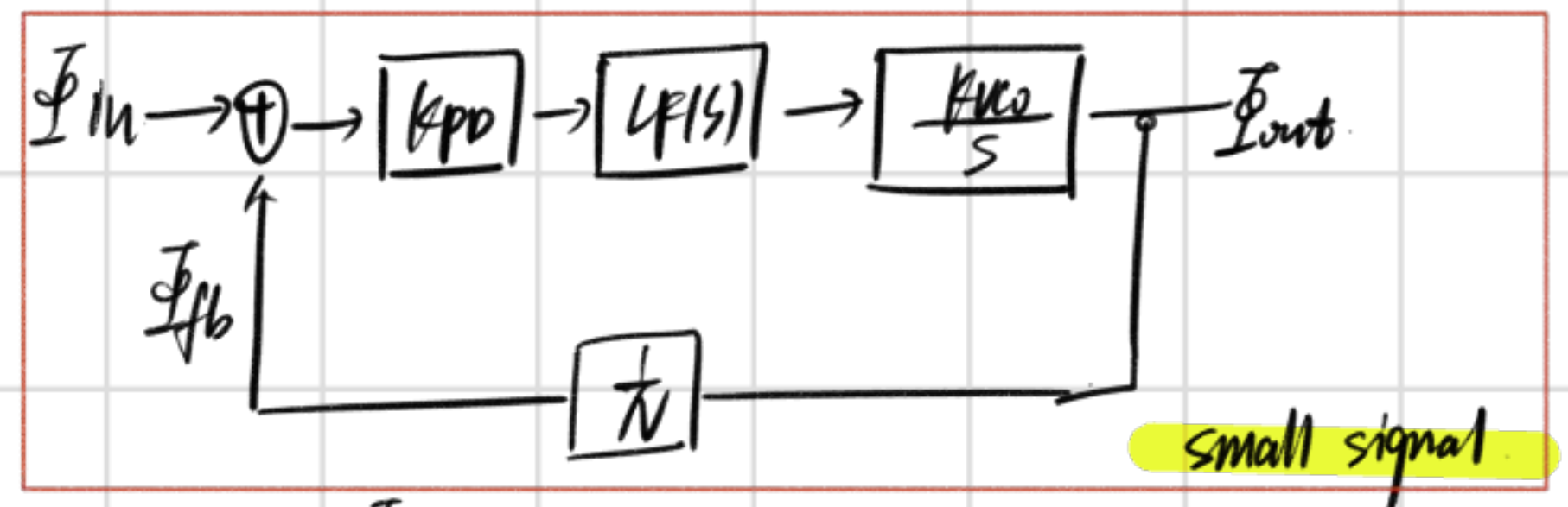
$$V_{out} = \cos(N \cdot W_{in}t)$$

- V_{fb} should have same frequency as V_{in} .
 - Compare $\frac{W_{out}}{N}$ with W_{in} and loop will enforce $\frac{W_{out}}{N} = W_{in}$.
- $$V_{fb} = A_1 \cos\left(\frac{W_{out}}{N} \cdot t\right)$$

Locking Condition:

$$\begin{cases} W_{out} = N W_{in} \\ \frac{d\phi_{er}}{dt} = 0 \end{cases}$$

$$\phi_{er} = W_{in}t - \int W_{fb} \cdot dt = W_{in}t - \int \frac{W_{out}}{N} \cdot dt = \phi_{in} - \frac{1}{N} \phi_{out}$$



$$\phi_{fb} = \frac{\phi_{out}}{N}$$

$$LG(s) = K_{pd} \cdot Lf(s) \cdot \frac{K_{vco}}{s} \cdot \frac{1}{N}$$

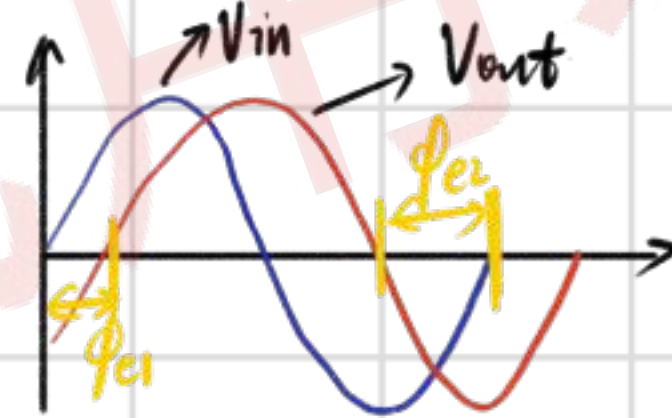
Analog Phase Error Detector



$$V_{in} = \sin(W_{in}t) = \sin(Wt + 0)$$

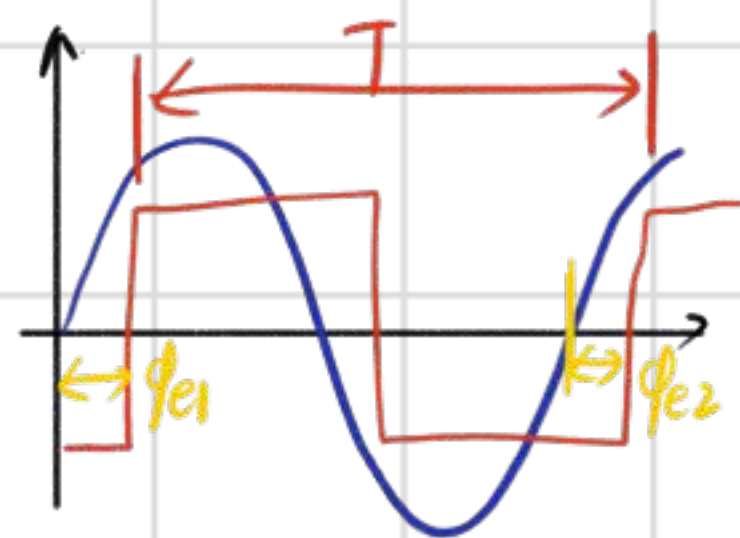
$$V_{out} = \cos(W_{out}t) = \sin(Wt - \phi_{er})$$

Phase Detector:



If frequency is the same:

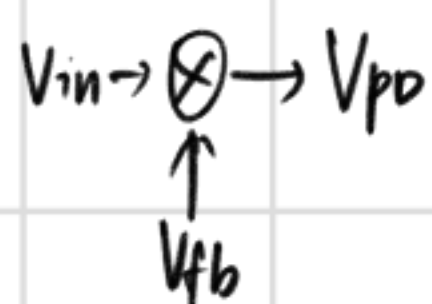
$$\phi_{in}(t) - \phi_{out}(t) = Wt - (Wt - \phi_e) = \phi_e$$



零点横坐标的间隔对应相位差: ϕ_{er}

$$\phi_{er} = 2\pi \cdot \frac{\Delta t}{T}$$

I. Multiplier or Mixer based PD

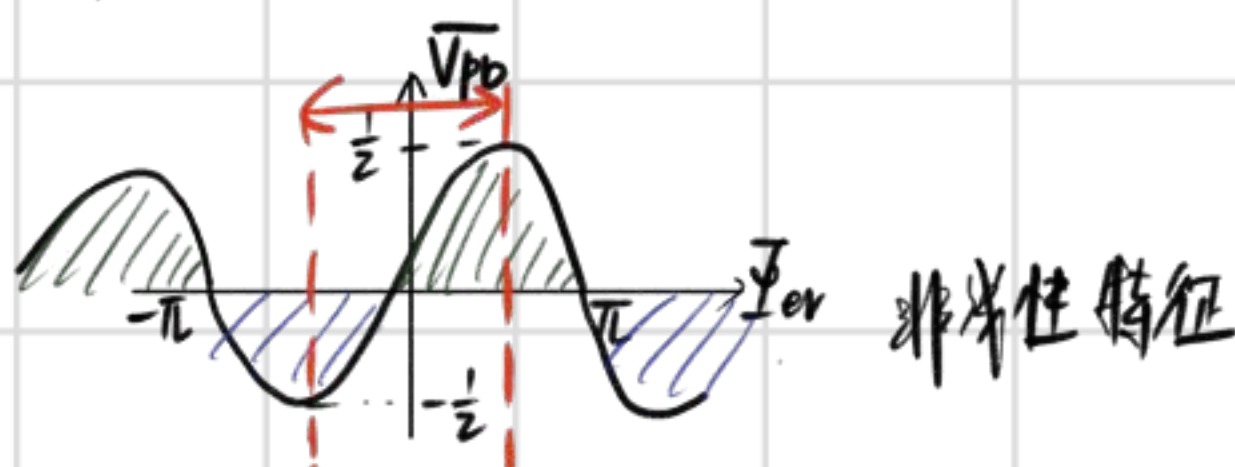


$$V_{in} = \sin(Wt + \phi_{in}(0))$$

$$V_{out} = \cos(Wt + \phi_{out}(0))$$

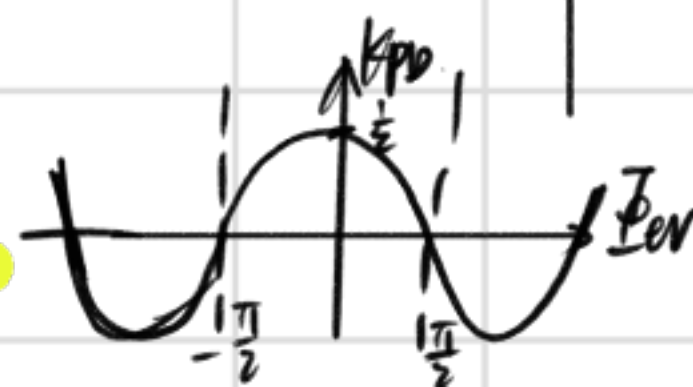
$$V_{pd} = \frac{1}{2} \left(\sin(2Wt + \phi_{in}(0) + \phi_{out}(0)) + \sin(\phi_{er}) \right)$$

Average of PD o/p, $\bar{V}_{pd} = \frac{1}{2} \sin(\phi_{er})$

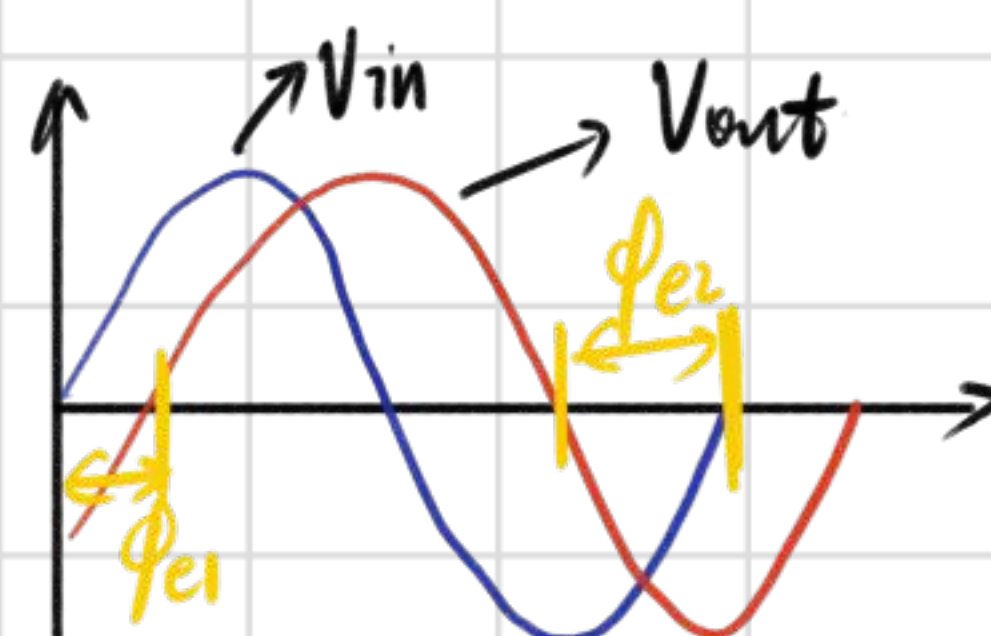
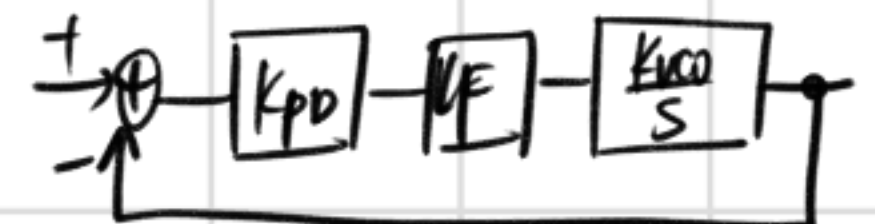
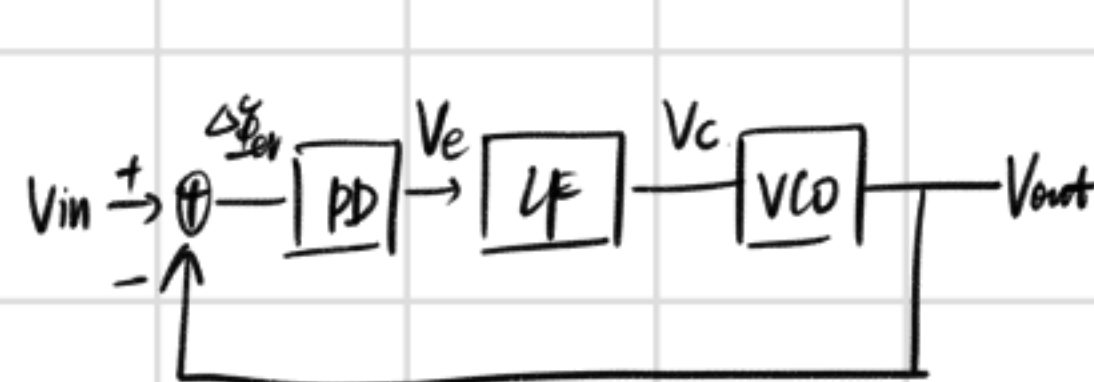


Linear Range (Monotonous Range): $\pm \frac{\pi}{2}$

$$\text{Gain of PD: } K_{pd} = \frac{d(\bar{V}_{pd})}{d\phi_{er}} = \frac{1}{2} \cos(\phi_{er})$$



Problem!!



If $\Delta\phi_{er} > \pi$, $V_{CO} \downarrow$,
go away from the locking!!

丧失线性特征.